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Energy-aware multi-robot exploration and coverage in fragmented unknown environments using collaborative reinforcement learning

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ABSTRACT

Navigating disaster-stricken environments presents significant challenges for autonomous multi-robot exploration, primarily due to extreme topological fragmentation characterized by an abundance of obstacles, narrow passages and numerous dead ends. Traditional multi-robot exploration strategies based on classical frontiers or static region partitioning, prove inadequate in these settings, leading to critical deficiencies in collaborative stability and energy safety, and a drastic reduction in overall exploration efficiency as robots frequently become stalled or disoriented. Therefore, this paper proposes the CERES-Q (Collaborative Energy-aware Reinforcement Exploration System with Q-learning), a novel closed-loop multi-robot exploration system for fragmented post-disaster scenarios that integrates a comprehensive "perception-decision-planning-execution-return" framework. This system leverages a phase-aware frontier clustering and allocation (CP-AFC) module, a dynamic task transfer (DTT) module, a safe return switch (SRS) with a safety Double Q-learning (SDQ) strategy to ensure reliable and efficient exploration. Comparative experiments on maps with varying degrees of fragmentation show that, compared with traditional multi-robot exploration methods, CERES-Q improves coverage efficiency by about 20% and reduces energy consumption by about 25%; compared with the recent Co-Explore method, it achieves about 10% better performance in highly fragmented environments. The advantages are particularly pronounced in scenarios with narrow passages and multiple dead ends. The results show that the CERES-Q is the key to coping with unknown and fragmented terrain after disasters and achieving reliable exploration and high-quality mapping; at the same time, the modular implementation of the framework facilitates expansion to larger teams and actual deployment.

1. Introduction

Post-disaster Search and Rescue (SAR) operations are intrinsically time-critical missions conducted in profoundly hazardous and dynamic environments. Structurally compromised settings, rapidly evolving topological layouts, and the imperative for rapid victim localization and extraction jointly amplify operational complexity and urgency. Crucially, secondary collapse risks and vast informational unknowns regarding the stability and content of the scene impose stringent constraints on the safety, scalability, and duration of human-led operations. This inherent volatility and risk therefore necessitate autonomous robotic systems capable of reliable rapid exploration, high-fidelity mapping, and cooperative mission execution with minimal reliance on *a priori* environmental knowledge.

The topological domain encountered in post-disaster environments poses an exceptionally challenging task for autonomous multi-robot exploration, fundamentally differing from controlled, structured settings such as indoor or warehouse environments. These operational theaters are defined by severe fragmentation, where navigable free space is heterogeneously interspersed with massive debris fields and collapsed structures, forming a complex network characterized by narrow, constricted corridors and a prevalence of low-connectivity dead-ends. This highly non-Euclidean spatial connectivity critically undermines the efficiency, completeness, and algorithmic reliability of conventional robotic navigation and path-planning methodologies. Real-time exploration and mapping in this hazardous context are subject to three interacting primary technical constraints that collectively motivate the need for novel research. First, the challenges begin with extreme informational entropy: catastrophic structural changes render any pre-disaster mapping data obsolete, creating an environment of profound informational uncertainty. Consequently, the autonomous system must execute Simultaneous Localization and Mapping (SLAM) in real time from a completely unknown initial state, which requires robust and adaptive perception-driven frameworks [1]. Second, the confluence of high obstacle

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density, physically restrictive narrow passages, and discontinuous free space severely limits robot reach and mobility, frequently creating localized topological traps. Simple preplanned coverage patterns (e.g., Boustrophedon or spiral decomposition) become ineffective, often inducing computationally expensive redundant looping, excessive detours, and oscillatory behaviors that consume substantial time and energy for marginal coverage gains [2]. Finally, exploration is strictly governed by finite onboard energy reserves, a constraint further exacerbated by fragmented layouts that demand repeated revisits and circuitous traversals to connect disconnected areas, accelerating power depletion [3]. Therefore, the core exploration strategy must be formulated as an explicit multi-objective optimization problem: the primary objective is to maximize coverage efficiency per unit energy expended, while the critical constraint is to guarantee the feasibility and successful execution of a safe return trajectory to the designated base station or charging depot. Successfully integrating these constraints—uncertainty, fragmented topology, and strict energy-safety mandates—into a unified and resilient exploration architecture is essential for sustainable and reliable autonomous operations in high-stakes post-disaster scenarios.

In recent years, researchers have made significant progress in two main directions. On the one hand, cooperative multi-robot exploration has advanced frontier processing, task allocation, and coordination mechanisms to reduce redundant traversal and mitigate interference in complex unknown environments [4–6]. On the other hand, energy-aware exploration/coverage and scheduling studies increasingly incorporate battery dynamics, heterogeneous depletion, and explicit recharging/return constraints to support operational safety and persistence [7–10]. However, in highly fragmented topologies dominated by narrow bottlenecks, collaborative stability and energy safety become tightly coupled in practice: instability at chokepoints can trigger repeated detours, idle waiting, and oscillatory goal switching, rapidly eroding the residual energy margin required to guaranty a safe return. Existing methods still tend to treat these aspects as separate layers. Coordination optimization typically does not explicitly consider the feasibility of returning in asymmetric battery states, while energy safety is usually imposed individually on each robot after coordination decisions [11,12].

In addition, traditional methodologies, encompassing both single-robot full-coverage planning and classic boundary- or region-based multi-robot partitioning strategies, are fundamentally unsuitable for the complexities of fragmented post-disaster topologies [13]. While such strategies can yield rapid initial coverage in conventional environments, they exhibit critical failures in fragmented terrain [14]. In particular, they suffer from three major drawbacks: algorithmic instability that causes vacillation among competing frontier objectives, frequent collaborative interference at critical bottlenecks, and a systemic neglect of energy-efficient and safe navigation heuristics. Collectively, these deficiencies undermine mission robustness and often lead to robot stalls, entrapment, or mission-critical energy depletion. Pure Reinforcement Learning (RL) methods, while potentially adaptive, face severe scalability challenges in this domain. Training safety-critical exploration behaviors is difficult within the vast state-action space of an unknown, topologically fragile environment, and the diversity of non-deterministic structural variations and debris configurations in post-disaster scenarios makes effective offline training computationally intractable and poorly generalizable. Consequently, RL is often better suited as a high-level guide for macro-action selection rather than as a comprehensive low-level controller for reliable path planning and frontier selection. The need, therefore, remains for a hybrid architecture that combines the exploration efficiency of multi-robot strategies with the robustness and safety guarantees lacking in both traditional and pure RL approaches.

To address these critical challenges in fragmented post-disaster environments, this paper introduces CERES-Q (Collaborative Energy-aware Reinforcement Exploration System with Q-learning)—a novel closed-loop multi-robot exploration system. The architecture is explicitly designed to enhance robustness, efficiency, and safety by enforcing a complete and continuous “perception–decision–planning–action–return” operational cycle. Here, “closed-loop” denotes not only the mission-level exploration-to-return cycle, but also real-time feedback among the core modules: CP-AFC produces temporally stable frontier candidates and ownership cues that shape SDQ macro-decisions; SDQ selects macro-actions (e.g., continue pursuit, switch/assist, or return) using geometric evaluations together with safety signals; DTT updates region boundaries/assignments based on unknown-rate and battery quotas, thereby reshaping subsequent CP-AFC frontier ownership; and SRS continuously evaluates return feasibility and can trigger safety overrides that prompt immediate re-planning and policy re-evaluation.

To explicitly connect the three interacting primary technical constraints with our four research highlights, CERES-Q couples perception, coordination, decision-making, and safety within a single operational loop. Under extreme informational entropy, CP-AFC stabilizes frontier representation and ownership, DTT tracks residual workload via region unknown-rate, and SDQ adapts macro-decisions to an evolving map. In fragmented topologies with bottlenecks and topological traps, CP-AFC reduces contention-driven oscillations and DTT enables cross-region assistance and boundary adjustments, providing SDQ with stable coordination options to avoid stalls. Under finite energy constraints, SRS enforces conservative safe-return feasibility with fallback guaranties, while DTT and SDQ incorporate battery quotas and safety-aware actions (e.g., return or postponement) to prevent energy-depletion failures. The framework achieves this through the tight coupling of four distinct, collaboratively engineered components:

1. Phase-aware Frontier Clustering and Allocation (CP-AFC): This module moves beyond conventional static frontier selection by extracting temporally stable representative frontiers. It utilizes a phase-aware metric to modulate clustering, ensuring ownership consistency. This design effectively suppresses target contention and allocation fluctuations that typically lead to interference and wasted effort at topological bottlenecks.
2. Dynamic Task Transfer (DTT) Module: The DTT actively manages the collaborative mission by continuously monitoring and fusing two critical metrics: the region unknown rate (quantifying remaining exploration work) and battery quotas (quantifying available operational capacity). This fusion triggers intelligent inter-region switching and dynamic boundary adjustments, ensuring sustained load balancing across the team and facilitating efficient cross-region assistance when a partner is stalled or depleted.
3. Safe-Return Switch (SRS): This component establishes a multi-layered safety net for energy-critical operations. It integrates a dynamic energy consumption model with conservative return planning—where unknown regions are conservatively treated as obstacles to preemptively calculate the return cost. If conservative planning fails due to new discoveries, it activates a breadcrumb fallback mechanism that leverages historically traversed, verified corridors to guarantee a safe return trajectory.
4. Safety Double Q-Learning (SDQ): A decentralized Double Q-learning approach is employed for strategic, high-level macro-action selection. This policy guides crucial non-geometric decisions, including: continuing frontier pursuit, initiating region switching, offering partner assistance, or executing a safe return or postponement maneuver. Geometric feasibility for all resulting paths is ensured by an underlying A* planner.

The subsequent sections of this paper are structured as follows: we first provide a comprehensive review of related work; we then detail the mathematical formulation and integration of the proposed CP-AFC, DTT, SDQ and SRS modules; this is followed by a presentation of the experimental methodology, comparative results, and an ablation study validating the contribution of each component; finally, we discuss the system’s limitations and outline future research directions concerning scalability to larger robot teams and physical deployment.

2. Related work

Multi-robot autonomous exploration in inherently hazardous, unknown post-disaster environments necessitates the simultaneous satisfaction of multiple, often competing, operational constraints including coverage efficiency, collaborative stability, and energy safety. A significant gap in existing literature is the tendency to address these requirements in isolation [15]. For instance, much of the research dedicated to multi-robot task allocation and collaboration focuses primarily on optimizing for spatial efficiency, specifically, minimizing redundant exploration and mitigating inter-robot conflicts at bottlenecks. This specialized focus frequently leads to a critical oversight, namely the omission of real-time onboard power constraints, which are paramount in ensuring mission completion and guaranteeing the safe retrieval of valuable assets from time-critical, energy-limited environments. This demonstrates the critical need for an integrated system architecture that holistically optimizes for both collaborative performance and safety-critical resource management.

Although deep reinforcement learning (DRL) techniques often surpass traditional heuristic algorithms in achieving superior coverage completeness in controlled or familiar domains [16], their practical application is limited by poor generalization capabilities and insufficient adaptability when deployed in entirely unknown and highly complex, fragmented post-disaster environments [17]. Concurrently, specialized energy-constrained exploration methods have introduced critical components like dynamic "return-to-base" policies and robust battery modeling; however, these approaches typically operate in isolation and lack the deep coupling necessary to effectively integrate energy state with multi-robot task allocation mechanisms. Consequently, relying on any single technological paradigm, be it DRL, heuristics, or isolated energy modeling is demonstrably insufficient to achieve the required trifecta of reliability, efficiency, and safety in fragmented, unknown post-disaster scenarios. This necessitates a systematic review and synthesis of the technological evolution across these three critical areas: exploration heuristics, advanced learning methods, and resource management, to establish the theoretical foundation for proposing a novel, closed-loop, multi-factor coupled exploration framework.

2.1. Multi-robot frontier allocation and coordination strategies

Early efforts in Multi-robot autonomous exploration were predominantly predicated upon a boundary-based paradigm, directing robotic agents toward frontiers which are the contiguous boundaries delineating mapped and unmapped regions as the primary navigation objective. [18]. The rationale was to systematically drive incremental mapping and achieve areal coverage by continuously reducing the perimeter between known and unknown operational space. However, a significant drawback of these initial multi-robot implementations was the absence of explicit inter-robot coordination mechanisms. This lack of enforced resource allocation frequently led to multiple agents converging toward the same frontier boundaries, resulting in high levels of target contention. The consequence was a cascade of inefficiencies, including redundant traversals of already-known areas and severe diminution of overall network coverage efficiency. This fundamental limitation highlighted the necessity for sophisticated collaboration strategies to transition from individual localized efficiency to reliable, globally optimal multi-robot performance. To address this issue, subsequent research proposed centralized allocation methods. For example, Burgard et al. (2005) proposed a centralized explorer that assigns non-conflicting target frontiers to each robot, significantly reducing redundant coverage [19]. However, centralized approaches rely on reliable communication and pose a single point of failure risk if the central controller fails. To improve the robustness of the search system, some researchers have proposed distributed task allocation mechanisms that enable robots to make local decisions while maintaining collaboration. A prominent line is market-based allocation, where robots bid for frontier tasks and assignments are iteratively updated to maximize team utility [20]. Zlot et al. (2002) modeled multi-robot exploration as a market economy, where robots bid for frontier targets through an auction, gradually allocating tasks to maximize overall utility [21]. This distributed auction improves scalability and fault tolerance, but performance degrades significantly when communication is unreliable or tasks are tightly coupled [22]. Building on this, Otte et al. (2020) improved the distributed auction mechanism, ensuring robust exploration even in communication-constrained environments, by adopting a method to balance the load while reducing conflicts [23]. The MinPos algorithm (Bautin et al.,2012) explicitly accounts for the spatial distribution of robots to minimize the detection of overlap and limit crowding [24]. Colares et al. (2016) combined information gain with travel cost to prioritize frontiers that are informative and economical, thus reducing duplication and improving overall performance [25]. Furthermore, Yu et al. (2021) proposed an exploration method based on a multi-objective potential field, which dynamically adjusts objectives through an artificial potential field to prevent robots from congregating in the same area [26].

Collectively, these strategies raise coverage efficiency and reduce interference in many settings. However, as environments become highly unknown, cluttered, and fragmented (especially with narrow passages and dead-ends pervasive after major disasters), their performance often degrades due to frequent assignment oscillations at bottlenecks, imbalanced workload across disconnected subregions, and the absence of energy-aware guarantees. Therefore, a closed-loop collaborative framework is needed to closely integrate task allocation with energy and environmental feedback to improve the stability and efficiency of exploration.

2.2. Integrating deep reinforcement learning with hybrid control for adaptive exploration

In recent years, RL has been applied to high-level decision-making in robot exploration, with the aim of learning intelligent exploration policies from experience. Early attempts in this domain focused on single-robot exploration. For example, Niroui et al. (2019) trained a deep RL agent to enable a ground robot to navigate unknown cluttered environments for search and rescue [27]. Their learned policy outperformed the classical frontier and next-best-view methods in terms of coverage efficiency, demonstrating the potential of RL to capture exploration strategies that adapt to environment layouts. However, applying RL to multi-robot systems is non-trivial due to the exponential growth of state and action spaces. A straightforward extension of single-robot RL to multi-robot (by treating the entire team as one combined agent) faces scalability issues and the non-stationarity problem, where each robot's policy sees other learning robots as part of a changing environment.

To handle multi-agent scenarios, researchers have turned to Multi-Agent RL (MARL) frameworks and centralized training with decentralized execution (CTDE) paradigms. Under CTDE, robots are trained together with access to global information (centralized training), but each executes its own policy using local observations during deployment (decentralized execution) [28]. In 2022, Deng et al. used Multi-Agent Deep Deterministic Policy Gradient (MADDPG) under a CTDE framework for multi-robot exploration in unknown environments, employing global information during centralized training while each robot executed its policy from local observations [29]. Zhang et al. (2022) uses the Hierarchical-Hops Graph Neural Networks (H2GNN) to perform exploration-task allocation in two stages: a coarse high-level region assignment followed by a fine-grained frontier selection, which leads to more efficient multi-robot cooperation. [30]. The macro-action RL framework proposed by Tan et al. (2023) reduces training complexity and improves inter-robot collaboration by abstracting high-level actions [31]. Regarding RL algorithm improvements, Su et al. (2023) combined Voronoi partitioning with deep Q-learning to achieve structured region allocation and local optimization [32]. The BiLSTM-D3QN

proposed by Wang et al. (2025) improves convergence stability and reduces overestimation through LSTM memory and Double-DQN mechanisms [33].

Beyond exploration-task policies, recent studies have also improved deep reinforcement learning for local path planning in obstacle-dense and uncertain environments. Xue et al. proposed a bidirectional obstacle-avoidance enhancement for DDPG to improve mobile-robot path planning in unknown dynamic scenes, aiming to strengthen collision avoidance and navigation reliability under complex obstacle interactions [34]. Moreover, Xue et al. introduced an action-curiosity-based DRL algorithm for path planning in nondeterministic environments, where curiosity-driven exploration and adaptive reward adjustment are used to improve learning efficiency and policy robustness [35]. In 2025, Xue et al. developed an improved DDPG method by improving the evaluation of path planning decisions in high-density environments, reporting better navigation performance and convergence behavior under cluttered conditions [36]. These works demonstrate the rapid progress of continuous learning-based control for challenging local planning problems. However, they primarily focus on single-robot navigation and do not explicitly address multi-robot frontier coordination, collaborative stability around bottlenecks, or energy-safe return guarantees, which are central in fragmented post-disaster exploration and motivate our closed-loop coupled design.

Despite recent progress, RL for multi-robot exploration still faces fundamental challenges in scalability, safety, and completeness.The state-action space grows rapidly with team size and environmental complexity, while multi-agent nonstationarity further complicates training. Moreover, pure RL policies typically lack mission-level guarantees (e.g., full coverage or energy-safe homing). Furthermore, generalization to new environments often requires substantial offline training or environment-specific retraining, with persistent limitations in sample efficiency and sim-to-real transfer. Consequently, recent work increasingly adopts hybrid control: constraining RL to high-level policy selection while delegating geometric feasibility and safety constraints to rule- or planner-based modules, thereby forming an interpretable, safety-aware closed loop [4]. This rules-plus-RL paradigm strikes a practical balance between adaptivity and robustness in complex unknown environments and is emerging as a strong direction for multi-robot exploration.

2.3. *Modeling and guaranteeing safe return in energy-constrained exploration*

Beyond task coordination and intelligent decision-making, the stringent constraints imposed by finite onboard energy reserves represent a critical operational limitation in autonomous exploration missions. Traditional exploration frameworks typically adopt a passive reactive threshold strategy, where robots are only triggered to return to base when the State-of-Charge (SOC) drops below a predetermined, fixed level. This approach is inherently high-risk, as demonstrated by early work such as Latif et al. (2020), which integrated a simple remaining-battery trigger into conventional coverage algorithms [37]. While providing a rudimentary safeguard, this passive return mechanism fails to account for dynamic energy consumption rates or environmental complexity, significantly compromising mission reliability and asset safety.

Specifically, when the battery level falls below a threshold, the robot automatically returns home and resumes exploration after recharging. However, this approach is extremely crude. A threshold that is too high can result in reduced coverage or wasteful power, while a threshold that is too low can easily cause the robot to get lost in the map [38]. Therefore, accurate modeling of battery usage is crucial for such strategies. For example, in 2025, Pant et al. proposed a hybrid control framework for multi-robot coverage that combines motion planning with energy dynamics [39]. They modeled each robot's battery state of charge (SOC) as part of its state and defined modes such as exploration mode (energy depletion) and return-to-base mode. By designing protection conditions for mode switching, their approach ensures that robots have sufficient energy to return to base when switching to return mode, preventing them from being "abandoned" due to battery depletion. However, relying solely on SOC guards to estimate the "return trip power" while ignoring details like steering, U-turns, and congestion can easily underestimate energy consumption on narrow, multi-turn maps, especially when the return trip requires multiple turns [40]. In addition to incorporating power constraints, another strategy is to combine path planning with energy constraints. For example, in 2024, Giang et al. proposed the BWave algorithm, which incorporates energy constraints into the generation of the coverage path, enabling the robot to complete the coverage of the area and return home within a single battery cycle [41].

In addition, team-based energy strategies have emerged. Meng et al. (2021) revealed the energy consumption disadvantages of distributed coverage and pointed out that centralized approaches are more energy efficient, highlighting the need for energy efficient collaboration [42]. In 2022, Rocamora et al. proposed a tree-based emergency landing logic behavior to prevent drones from crashing when their battery suddenly drops [43]. Fink et al. (2023), drawing on the "breadcrumbs" concept from fairy tales, proposed to establish a backtracking path during Martian cave exploration by deploying small communication nodes [44]. In 2024, Datsko et al. published a proposal for energy-efficient multi-drone coverage mission planning that integrates energy constraints into the generation of coverage paths, enabling each drone to complete its assigned area and return safely within a single battery cycle [45]. However, existing methods often decouple energy management from task allocation, leading to a separation between return and collaborative optimization. In fragmented terrain, existing algorithms often underestimate the power required for return when encountering unexpected detours or dead ends, which poses a risk. Therefore, a closed-loop energy coordination system is needed that tightly integrates task allocation, energy status, and return planning to ensure that the robot maximizes coverage efficiency while ensuring return safety.

In summary, the trajectory of multi-robot exploration research from simple greedy frontier algorithms to sophisticated utility-driven and market-based allocation mechanisms, alongside the integration of DRL policies still fails to provide robust solutions for highly fragmented topologies. These advanced methods are consistently plagued by pervasive issues such as target oscillation at bottlenecks, uneven workload distribution across disconnected subregions, and the critical problem of energy fragmentation (i.e., the inability to guarantee safe return in complex terrain). To overcome these systemic limitations, this paper proposes the CERES-Q system. Our core contribution is the integration of collaboration and energy constraints within a unified, continuous "perception-decision-planning-execution-return" control loop, which is essential for ensuring both reliable and efficient autonomous exploration in unknown, complex post-disaster environments.

3. Methodology

To enable robust and comprehensive autonomous operation within complex and unknown post-disaster topologies, this paper introduces the CERES-Q system (Collaborative Energy-aware Reinforcement Exploration System with Q-learning). The architecture implements a continuous operational cycle spanning from initial deployment through safe return. Collaboration and efficiency are realized through the integration of the Collaborative Phase-aware Frontier Clustering (CP-AFC) module and the Dynamic Task Transfer (DTT) module, which together manage dual-robot task assignment, mitigate contention, and ensure sustained workload balance. Crucially, a Safety Double Q-Learning (SDQ) mechanism is employed as a high-level behavioral policy, which is robustly trained offline to guide critical collaborative decisions and macro-actions. Finally, mission integrity is guaranteed by a Safe-Return Switch (SRS), which dynamically monitors and triggers the return phase based on real-time energy consumption modeling, thereby replacing unreliable fixed thresholds. This design establishes a continuous, tightly-coupled

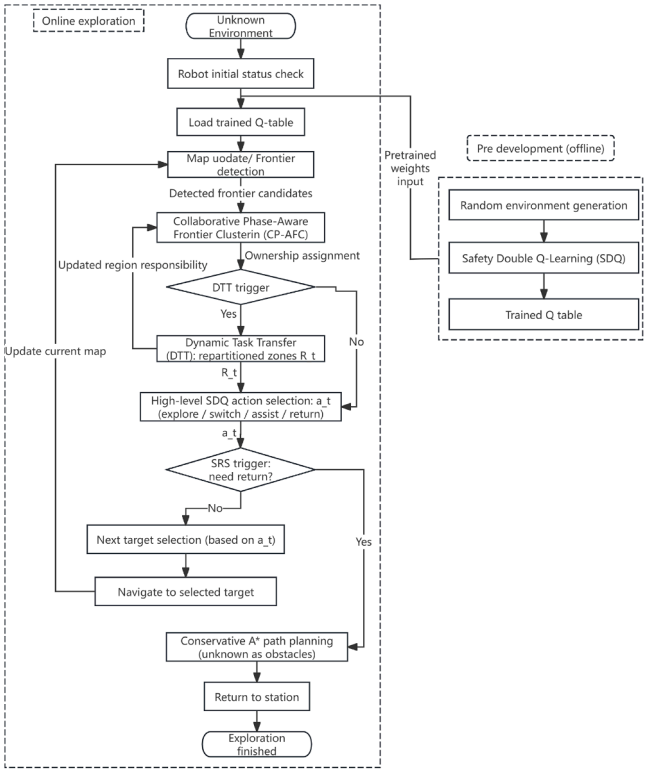


Fig. 1. Operational flowchart of the closed-loop multi-robot exploration system.

"exploration-collaboration-decision-return" closed-loop process. Fig. 1 provides a visual overview of this complete architectural cycle.

Fig. 1 outlines the closed-loop execution of CERES-Q and the interfaces between CP-AFC, DTT, SDQ and SRS. In each decision cycle, the robot updates the map M_t and extracts raw frontier candidates F_{raw}. CP-AFC then performs phase-aware clustering to generate stable representative frontiers and temporary ownership cues $\{F_{\text{rep}}, O_{\text{rep}}\}$. This mechanism reduces contention and suppresses oscillatory behaviors near bottlenecks by maintaining short-horizon consistency in frontier ownership. The DTT module is event-triggered and continuously monitors the unknown rate of the region, which reflects the remaining exploration workload, together with the battery quota, which reflects the available operational capacity. When these signals indicate a significant imbalance, DTT updates the region responsibility map R_t. The updated R_t is sent back to CP-AFC as a mask of the region and as the prior ownership, constraining the admissible frontier set for each robot and reshaping the frontier allocation in subsequent cycles.

SDQ outputs a macro-action

$$a_t \in \{\text{explore}, \text{switch}, \text{assist}, \text{return}\}. \tag{1}$$

The target selection interface then converts a_t into a concrete goal g_t by combining a_t with $\{F_{\text{rep}}, O_{\text{rep}}\}$ and the current R_t. When a_t = explore, the goal is selected from representative frontiers within the robot's assigned region. When a_t = switch or a_t = assist, the target is selected in the target region determined by R_t, such as a boundary-entry point or a representative frontier within that region. When a_t = return, the goal is set to the location of the base station. The low-level planner then computes a feasible path. For exploration-related actions, $\tilde{M}_t$ is the current traversable map. For safety-critical return, $\tilde{M}_t$ is conservative and treats unknown cells as obstacles. SRS continuously evaluates return feasibility with the highest priority. If return becomes unsafe, SRS overrides the current decision by imposing a_t = return, resetting g_t to the base station and triggering immediate replanning. If the conservative return path becomes infeasible, SRS activates a breadcrumb backtracking fallback along previously verified corridors.

Table 1 summarizes the recommended ranges of key parameters in CERES-Q together with the rationale for their selection. These ranges are determined using a layered procedure that integrates domain constraints, algorithmic stability considerations, and controlled validation runs, with the goal of maintaining stable closed-loop behavior in post-disaster fragmented topologies.

All parameters used in the comparative evaluation are fixed prior to the main experiments and remain unchanged across all test cases, ensuring the reproducibility and fairness of the experimental setup. Since the parameter set spans a large combinatorial space, we constrain and calibrate key parameters using representative validation scenarios and report practical intervals that support stable execution and avoid obvious degraded behaviors. The resulting ranges are intended to serve as a reproducible default reference for future users, thereby reducing reliance on case-specific tuning.

Among the parameters listed in Table 1, the phase weight w_θ in CP-AFC and the cooldown step size H in DTT have the strongest influence on observed behavior. The phase weight w_θ regulates the strength of phase-consistent ownership in frontier allocation. Increasing w_θ generally improves the stability of short-horizon ownership and reduces contention-driven oscillations near bottlenecks, while overly large values may slow adaptation when the exploration phase changes or when the local topology evolves rapidly. The size of the cooldown step H limits the frequency at which DTT can trigger repartitioning. Smaller values improve responsiveness and accelerate load adjustments, but may increase the likelihood of frequent responsibility switching in highly fragmented terrains. Larger values improve coordination stability and reduce oscillatory switching, but may delay the effectiveness of cross-region assistance and load balancing.

3.1. Collaborative phase-aware frontier clustering (CP-AFC)

In environments lacking a priori mapping information, achieving comprehensive exploration typically relies on established methods such as boundary-based frontier exploration or various forms of systematic search strategies. Boundary-based approaches direct autonomous agents toward the perimeter between known and unknown space (the frontier), thus driving incremental mapping. Conversely, systematic search strategies (e.g., coverage paths) guarantee thorough inspection of the environment but often require geometric decomposition or pre-knowledge of free space. While both paradigms aim for maximal coverage, they frequently encounter limitations regarding efficiency, especially when dealing with complex, non-Euclidean topologies. However, in post-disaster scenarios, the dense distribution of obstacles resulting from significant environmental changes could make traditional geometric boundary grouping methods unreliable. This unreliability may result in the robot getting stuck or lost.

The left panel of Fig. 2 illustrates a critical failure mode inherent in traditional frontier clustering methodologies. These algorithms rely solely on Euclidean geometric distance between frontier points, neglecting crucial information regarding topological connectivity imposed by environmental obstacles. As depicted within the red dashed circle, this geometric-only approach leads to the erroneous grouping of frontier segments located on opposite sides of a physical obstruction, such as a thin wall or narrow passage. This flawed clustering results in the selection of a single representative frontier (F^*) that is geometrically central but topologically inaccessible from the robot's current position (Robot A). The robot is then compelled to repeatedly attempt to navigate towards this inaccessible point across the obstacle, leading to ineffective path planning, repeated localized wandering, and the formation of a dead-loop cycle, severely degrading exploration efficiency.

To mitigate the critical failures of traditional methods within fragmented topologies, we propose the CP-AFC mechanism. This novel approach integrates a direction-enhanced boundary clustering method with a dynamic multi-robot allocation strategy. The core innovation lies in utilizing both geometric proximity and topological phase congruity

Table 1
Unified Glossary and Parameter Set for the Proposed Closed-Loop Exploration System. This table defines all mathematical symbols used in the system's core equations, including recommended operating ranges and parameter types for the CP-AFC, DTT, SDQ, Safe Return Switch modules.

Parameter	Recommended Range / Type	Definition
Collaborative Phase-Aware Frontier Clustering (CP-AFC)		
N_4	{4, 8} (enum)	Neighborhood connectivity used in frontier/open-ness terms (paper uses 4).
m_i	integer (odd, coprime across i)	Ring-phase period for robot i in $\varphi_i(\cdot) = \mathrm{mod}(\|p_f - p_i\|_1, m_i)$.
w_θ	0.4∼1.0 (real)	Weight of the phase-difference term in the augmented distance d_{aug}.
B	4∼16 (integer)	Number of angular sectors used in the directional histogram (entropy H_m).
W (window size)	radius 2–4 (cells)	Local window behind representative for unknown/free ratios
λ_1	0.3∼1.2 (real)	Weight on the free-ratio term in the EV numerator.
λ_2	0.1∼0.8 (real)	Distance-cost scaling in the EV denominator (with d_M).
λ_3	0.2∼0.8 (real)	Directional-freedom boost coefficient in EV.
β	0.2 ∼ 1.0 (real)	Weight on heading-alignment term in ownership cost
Dynamic Task Transfer (DTT)		
N	≥ 2 (integer)	Number of robots (this paper uses $N=2$).
H	3∼15 (steps)	Cooldown: minimum steps between two repartitions in trigger $\mathcal{T}_{\mathrm{DTT}}$.
P	3∼10 (steps)	Period: repartition is evaluated only on ticks with $\tau \bmod P = 0$.
σ_{stop}	0.05∼0.20 (real)	Remaining-unknown ratio gate in trigger; below it, repartition is disabled.
b_0	1∼3 (columns)	Fixed soft-band half-width around the cut c^* in $B = \{ k : \|k - c^*\| \le b_0 \}$.
Safety Double Q-Learning (SDQ)		
Q	4∼8 (integer)	Number of bins for teammate heading/quadrant in $q_\theta \in \{0, \dots, Q-1\}$.
w_c	5∼20 (real)	Weight on coverage increment in reward r_t.
w_e	0.05∼0.30 (real)	Weight on energy penalty in r_t.
w_o	3∼10 (real)	Weight on overlap/conflict penalty in r_t.
w_a	1∼6 (real)	Weight on cooperation/transfer bonus R_{eff} in r_t.
w_d	5∼12 (real)	Weight on dead-end/exception penalty in r_t.
w_{r}	8∼20 (real)	Weight on successful docking (terminal bonus) in r_t.
w_{w}	0.5∼3 (real)	Weight on idling/wandering penalty in r_t.
η	0.01∼0.05 (real)	Improvement margin for $R_{\mathrm{eff}} = \mathbf{1}\{\Delta\mathrm{trans} < -\eta\}$.
γ	0.90∼0.99 (real)	Discount factor in the Double Q-learning update.
Safe Return Switch (SRS)		
k_d	calibrated ≥ 0 (real)	Energy per unit distance in $E^{\mathrm{safe}}_{\mathrm{return}} = (1+u)(k_d L_{\mathrm{return}} + k_\theta T_{\mathrm{turn}})$.
k_θ	calibrated ≥ 0 (real)	Energy per 90° turn in $E^{\mathrm{safe}}_{\mathrm{return}}$.
u	0.10∼0.30 (real)	Safety margin multiplier in return-energy estimation.
$\mathrm{Cov}_{\mathrm{min}}$	0.80∼0.95 (real)	Minimum coverage threshold in the return trigger.

to group frontiers, thereby significantly enhancing boundary grouping stability. By minimizing the inclusion of topologically separated frontier points into a single cluster, the CP-AFC mechanism effectively suppresses target contention and reduces localized oscillation, ensuring that robots are guided to explore the map with greater efficiency and minimize inter-robot conflict in highly fragmented environments.

In an occupancy grid, each cell state is denoted by $M(i,j) \in \{0,1,2\}$, where 0 indicates an unknown state, 1 indicates a free space and 2 indicates that the cell is occupied. The 4-connected neighborhood of a cell is defined as $N_4(i,j)$. A cell (i,j) is recognized as a frontier if it is free and at least one of its neighbors in $N_4(i,j)$ is in an unknown state. Formally, the frontier set can be defined in Eq. (2).

$$F = \{ f = (i,j) \mid M(i,j) = 1 \ \wedge \ \exists (i',j') \in N_4(i,j) \ : \ M(i',j') = 0 \}. \tag{2}$$

Once the frontiers are identified, the frontier cells are clustered using a density-based clustering procedure. This approach augments the distance metric with anchor-relative ring-phase differences, which helps in grouping boundaries that are spatially close and phase consistent with respect to the robots. For each frontier f and robot i, a ring phase $\varphi_i(f)$ is defined from the Manhattan distance to the robot anchor,

Let N denote the number of robots in the team and index the robots by $i \in \{1, \dots, N\}$. Each robot i has a grid position (anchor) $p_i = (x_i, y_i)$ and a heading θ_i. Each frontier cell f has a grid coordinate $p_f = (x_f, y_f)$. Since the environment is represented on a 4-connected occupancy grid, we use the Manhattan distance $\|\cdot\|_1$ as the base metric in the following definitions.

$$\varphi_i(f) = \mathrm{mod}\big(\|p_f - p_i\|_1, \, m_i\big), \quad i \in \{1, \dots, N\}. \tag{3}$$

The increased distance between two frontiers f_a and f_b is expressed as

$$\begin{aligned} d_{\mathrm{aug}}(f_a, f_b) &= \|p_a - p_b\|_1 + \sum_{i=1}^{N} w_\theta \, \Delta\varphi_i(f_a, f_b), \\ \Delta\varphi_i(f_a, f_b) &= \min\Big(\big|\varphi_i(f_a) - \varphi_i(f_b)\big|, \ m_i - \big|\varphi_i(f_a) - \varphi_i(f_b)\big| \Big). \end{aligned} \tag{4}$$

where $p_a = (x_a, y_a)$ and $p_b = (x_b, y_b)$ represent the coordinates of f_a and f_b, p_i is the anchor position of robot i, m_i is the period of the ring, w_θ is a weighting factor, and $\Delta\varphi_i$ denotes the minimal difference in phase of the ring. Frontier cells are categorized into the same cluster when the augmented distance is below a predefined threshold. In Eq. (4), the summation is taken over all robots ($i = 1, \dots, N$), which means that the phase disagreement between two frontier cells is evaluated with respect to each robot anchor. In our experiments, $N = 2$, the definition is directly generalized to larger teams. This multi-anchor phase term helps keep two leading segments separated by obstacles in phase with the team, thereby reducing cross-obstacle clustering in fragmented maps.

When the method is applied to multiple robots, each cluster is dynamically assigned to one robot to prevent redundant exploration. The ownership cost for robot i to claim cluster m is defined as

$$\begin{aligned} O_{i,m} &= D_{i,m} + \beta \, \Delta\theta_{i,m}, \\ \Delta\theta_{i,m} &\equiv \mathrm{angdiff}\big(\theta_i, \ \Theta_m\big). \end{aligned} \tag{5}$$

In Eq. (5), $D_{i,m}$ measures the (normalized) travel effort from robot i to cluster m, while $\Delta\theta_{i,m}$ measures the mismatch of directions between the current direction of the robot θ_i and the principal orientation of the cluster Θ_m. We compute $\Delta\theta_{i,m}$ using an absolute angular difference and normalize it to $[0,1]$ so that $D_{i,m}$ and $\Delta\theta_{i,m}$ are commensurate. The weight $\beta \ge 0$ trades off distance-driven assignment versus heading-consistent assignment: smaller β makes ownership primarily distance

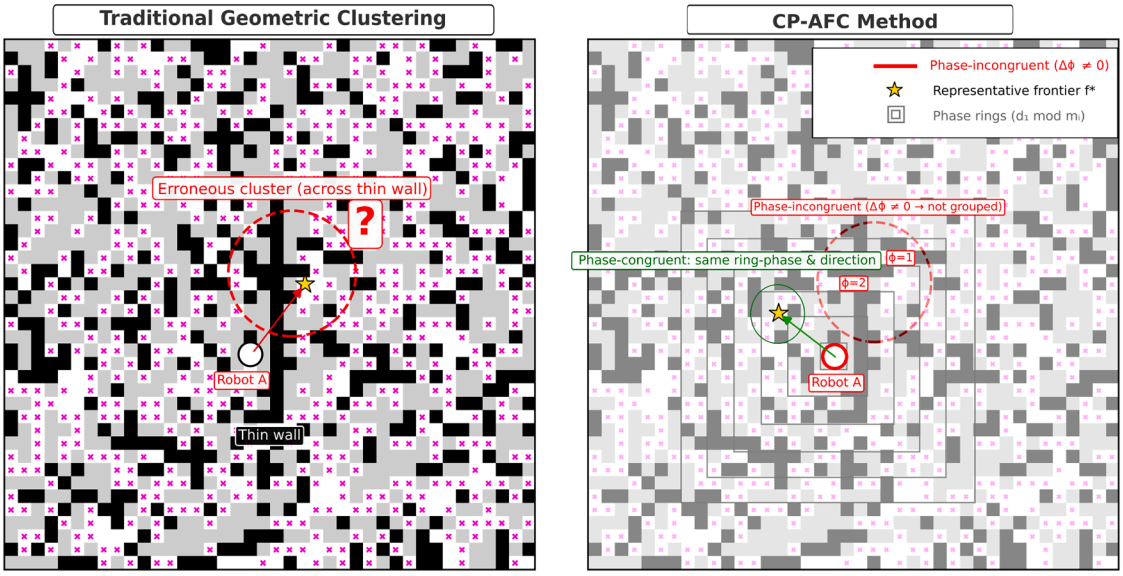


Fig. 2. Schematic diagram of the difference between our method and traditional methods. Left: geometric-distance clustering groups frontiers across a thin wall, yielding an inaccessible representative f^*. Right: CP-AFC introduces ring-phase and phase-congruency to prevent cross-obstacle grouping and stabilize cluster ownership.

-based (potentially increasing turning and head-on congestion near bottlenecks), whereas larger β favors clusters aligned with the current motion direction (reducing turning and crossover conflicts but potentially delaying off-heading yet informative frontiers). Therefore, the values in Table 1 are recommended to strike a balance between responsiveness and stability in fragmented maps.

In each assigned cluster, a single representative frontier is selected to serve as the exploration goal. To ensure stability and maximize information gain, this representative frontier must align in the predominant direction of the group Θ_m. As indicated in Eq. (6), among the candidates that satisfy this phase-lock condition, the one with the largest number of unknown cells in its local window is chosen.

$$f_m^* = \arg\max_{f \in C_m} W(f). \tag{6}$$

In Eq. (6), C_m denotes the set of candidate frontiers in cluster m, and $W(f)$ represents the number of unknown cells within a predefined local neighborhood behind the frontier f, serving as a proxy for its expected information gain.

As shown on the right of Fig. 2, CP-AFC introduces the ring phase ($d_1 \bmod m_i$) to distinguish frontier points (circled in red on the left) that are geometrically close but topologically separated by obstacles. Combined with the phase consistency criterion ($\Delta\varphi = 0$) and the robot orientation constraint, CP-AFC clusters only those frontier points with consistent phases and explicitly excludes those with inconsistent phases. This mechanism effectively prevents clustering across obstacles, preventing the robot from getting stuck or not exploring. To further evaluate the exploration value of different representative frontiers, the exploration value (EV) is introduced. This metric reflects the potential information gain, the local traversability, and the travel effort. Concretely, for a cluster m with representative frontier f_m^*, $\mathcal{W}(f_m^*)$ is defined as a fixed local window behind f_m^*, and the unknown ratio u_m and the free ratio ϕ_m in this window are calculated in Eq. (7)

$$\begin{aligned} u_m &= \frac{\left|\{\, n \in \mathcal{W}(f_m^*) \mid M(n) = \text{unknown} \,\}\right|}{\left|\mathcal{W}(f_m^*)\right|}, \\ \phi_m &= \frac{\left|\{\, n \in \mathcal{W}(f_m^*) \mid M(n) = \text{free} \,\}\right|}{\left|\mathcal{W}(f_m^*)\right|}. \end{aligned} \tag{7}$$

The directional degrees of freedom, $\chi_m \in [0,1]$, indicate the number of available free directions around f_m^* and the level at which these directions are distributed. They are calculated by multiplying the normalized angular entropy H_m with the local openness factor O_m.

$$\begin{aligned} \chi_m &= H_m \cdot O_m, \\ H_m &= -\frac{1}{\log B} \sum_{b=1}^{B} q_{m,b} \log q_{m,b}, \\ O_m &= \frac{\left|\{\, n \in N_4(f_m^*) \mid M(n) = \text{free} \,\}\right|}{4}. \end{aligned} \tag{8}$$

where $\{q_{m,b}\}_{b=1}^{B}$ is the normalized histogram of free cells in the B angular sectors in a neighborhood of f_m^*. The travel effort is measured by the Manhattan distance $d_M(p, f_m^*)$ from the robot's current position p to f_m^*, and EV_m is calculated in Eq. (9).

$$EV_m = \underbrace{\frac{u_m + \lambda_1 \phi_m}{1 + \lambda_2 d_M(p, f_m^*)}}_{\text{information gain vs. travel cost}} \cdot \underbrace{\left(1 + \lambda_3 \chi_m\right)}_{\text{directional freedom boost}}. \tag{9}$$

In Eq. (9), $u_m \in [0,1]$ represents the unknown ratio behind the representative boundary (representing information gain), $\phi_m \in [0,1]$ represents the free space ratio (representing local mobility), and $\chi_m \in [0,1]$ represents the directional degrees of freedom (representing the probability that the boundary points to a well-connected branch rather than a dead end). The coefficient λ_1 controls the strength of the supplementary effect of local mobility on information gain; λ_2 scales the distance penalty in grid units and determines the degree of discounting of the boundary; λ_3 strengthens the boundary with higher degrees of directional freedom to reduce traps and re-tracking in branches with low connectivity. The recommended ranges in Table 1 are to keep these normalization terms on a comparable scale and avoid degenerate behavior: an excessively large λ_2 will make the policy too short-sighted, while an excessively large λ_3 may over-prefer open regions and postpone the exploration of narrow branches necessary at high-coverage targets. The pseudo-code for CP-AFC is outlined in Algorithm 1.

3.2. Dynamic task transfer (DTT)

Dynamic Task Transfer (DTT) aims to achieve an adaptive balance between workload and energy consumption during multi-robot

Algorithm 1 CP-AFC (Phase-aware frontier clustering and selection).

Detect frontier set F from the occupancy grid and remove noisy candidates
for each frontier $f \in F$ and each robot $i \in \{1, \dots, N\}$ **do**
  Compute ring-phase $\varphi_i(f) \leftarrow \mathrm{mod}\big(\|p_f - p_i\|_1,\ m_i\big)$
end for
Run DBSCAN on F using augmented distance

$$d_{\mathrm{aug}}(f_a, f_b) = \|p_a - p_b\|_1 + w_\theta \sum_{i=1}^{N} \big|\varphi_i(f_a) - \varphi_i(f_b)\big|$$

for each cluster C_m **do**
  Assign owner $r_m = \arg\min_i \Big(D_{i,m} + \beta\, \Delta\theta_{i,m} \Big)$
  Form phase-aligned set $C_m^{\mathrm{align}} \subseteq C_m$ by keeping candidates consistent with the owner phase
  for each candidate $f \in C_m^{\mathrm{align}}$ **do**
    Compute $(u(f), \phi(f), \chi(f))$ and

$$EV(f) = \frac{u(f) + \lambda_1 \phi(f)}{1 + \lambda_2\, d_M(p_{r_m}, f)} \cdot \big(1 + \lambda_3 \chi(f)\big)$$

  end for
  Select cluster representative $f_m = \arg\max_{f \in C_m^{\mathrm{align}}} EV(f)$ and set $EV_m = EV(f_m)$
end for
Select global target $f = \arg\max_m EV_m$
return f and its owner r_m

exploration. Unlike traditional methods that rely on static partitioning or simple load balancing, DTT continuously monitors the number of unknown cells in each area and the remaining energy of each robot, dynamically adjusting the division of responsibility areas during exploration. This mechanism effectively prevents situations where one robot becomes overloaded while others remain idle, or where a robot is unable to complete its assigned area due to insufficient battery.

Concretely, DTT divides the discretized unknown grid map into n regions, and the number of unknown cells in subregion k ($k \in \{1, 2, \dots, n\}$) is denoted as $U^{(k)}$. During exploration, the counts of unknown cells are updated monotonically, which means that the state of cells can only transition from 'unknown' to either 'free' or 'obstacle' without being reverted or counted redundantly. This ensures that coverage accounting remains stable throughout the mission. Therefore, the number of unknown cells assigned to robot i can be calculated as

$$U_i \;=\; \sum_{k \in \mathcal{R}_i} U^{(k)}, \qquad i \in \{1, \dots, N\}. \tag{10}$$

where N is the number of robots (in this method, $N = 2$) and $\mathcal{R}_i$ is the set of subregions assigned to robot i. Meanwhile, the current workload ratio and the energy target ratio of robot i are

$$\alpha_i \;=\; \frac{U_i}{\sum_{j=1}^{N} U_j}, \qquad t_i \;=\; \frac{a_i b_i}{\sum_{j=1}^{N} a_j b_j}. \tag{11}$$

where b_i is the remaining energy of robot i and $a_i \in \{0, 1\}$ indicates whether robot i is in the exploration state. When a robot switches to 'return' mode ($a_i = 0$), its weight automatically vanishes, and the remaining robot will take over the workload.

As exploration progresses, the system periodically checks whether a repartition is needed so that regional boundaries can be adaptively adjusted. The trigger condition for DTT is shown in Eq. (12).

$$\mathcal{T}_{\mathrm{DTT}} \;\equiv\; \big[(\tau - \tau_{\mathrm{last}}) \geq H\big] \;\wedge\; \big[\tau \bmod P = 0\big] \;\wedge\; \left[\frac{U_{\mathrm{rem}}}{U_{\mathrm{init}}} > \sigma_{\mathrm{stop}}\right]. \tag{12}$$

where τ is the current step, τ_{last} the last transfer step, H is the cooling step, P is the repartition check period, U_{init} is the initial number of unknown cells, and $U_{\mathrm{rem}} = \sum_{k=1}^{n} U^{(k)}$ is the remaining unknown mass. The gate σ_{stop} prevents unnecessary transfers when only a small part of the environment is unexplored, making the benefits of rebalancing negligible.

Once the above condition is satisfied, the cumulative unknown ratio $C(c)$ is calculated to determine the new cut position, where C is defined under a vertical strip-based partitioning scheme.

$$C(c) \;=\; \frac{\sum_{k<c} U^{(k)}}{\sum_{k=1}^{n} U^{(k)}}, \qquad c \in \{1, \dots, n-1\}. \tag{13}$$

and select the new cut by minimizing,

$$L(c) \;=\; \Big| C(c) - t_1 \Big|, \qquad c^* \;=\; \arg\min_{c \in \{1,\dots,n-1\}} L(c). \tag{14}$$

Therefore, the left region $\{\, k < c^* \,\}$ matches *robot*1's energy share t_1, while the remainder is assigned to *robot*2.

To improve accessibility near the partition boundary, a fixed-width soft band centered at c^* is introduced. Within this band, cells are reassigned to the nearest robot instead of strictly following the hard cut.

$$\Delta c \;=\; b_0, \qquad \mathcal{B} \;=\; \{\, k \mid |k - c^*| \leq \Delta c \,\}. \tag{15}$$

where b_0 is the base half-width of the soft band. Cells in $\mathcal{B}$ are redistributed based on their proximity using the principle of proximity adsorption.

$$i^*(g) \;=\; \arg\min_{i \in \{1,2\}} d_M\big(g, p_i\big). \tag{16}$$

where, $i^*(g)$ indicates that the grid cell g is assigned to robot i, p_i is the current pose of robot i, and d_M is the Manhattan distance. DTT uses hard partitioning for the area outside the soft band and collaborative partitioning for the area inside the soft band, ultimately obtaining the current task partition $\hat{U}_i$ of the robot i,

$$\begin{aligned} \hat{U}_1 \;&=\; \sum_{k<c^*-\Delta c} U^{(k)} \;+\; \sum_{g \in \mathcal{G}_B} \mathbf{1}\big[\, i^*(g) = 1 \,\big], \\ \hat{U}_2 \;&=\; \sum_{k>c^*+\Delta c} U^{(k)} \;+\; \sum_{g \in \mathcal{G}_B} \mathbf{1}\big[\, i^*(g) = 2 \,\big], \\ \mathcal{G}_{\mathcal{B}} \;&=\; \{\, g \mid k(g) \in \mathcal{B} \,\}. \end{aligned} \tag{17}$$

To further illustrate the operation of the proposed DTT mechanism, Fig. 3 presents a sequence of snapshots (S0-S3) from a simulated two-robot exploration task.

At **S0**, both robots start with equal energy levels (100% each), so the cut line c^* is placed at the center, resulting in balanced regions. The soft band appears at the partition boundary, serving as a buffer to ensure that cells near the boundary are assigned to the nearest robot, thus avoiding rigid boundaries at the early stage. At **S1**, the energy changes to (80%, 90%), but the imbalance is still minor. The DTT trigger condition is met, and the cumulative unknown curve $C(c)$ deviates slightly from the target t_1. Consequently, c^* shifts slightly leftward, $t = [0.47, 0.53]$, indicating that DTT can smoothly adjust without excessive oscillation. At **S2**, the imbalance becomes more significant with (40%, 70%), and the target ratio is $t = [0.34, 0.66]$. The cut line c^* shifts clearly left, allocating more unexplored area to Robot 2. Since the soft band is closer to Robot 1, some boundary cells are still temporarily assigned to it, ensuring the partition's continuity and reachability. This highlights the role of the soft band absorption: while redistributing workload according to energy, it preserves smooth transitions across the boundary. At **S3**, Robot 1 drops to 5% energy and enters return mode ($a_1 = 0$). Its weight is gated to zero, and nearly all unexplored cells are reassigned to Robot 2. The soft band remains visible at the boundary, not for allocating tasks to Robot 1, but to maintain structural consistency and reduce unnecessary replanning. Moreover, if Robot 1 passes near the band during its return, it can still opportunistically update nearby cells.

The procedural outline of DTT is presented in Algorithm 2.

3.3. Safety double Q-Learning (SDQ)

Building on the foundational algorithm, this project introduces Safety Double Q-Learning (SDQ), combining reinforcement learning

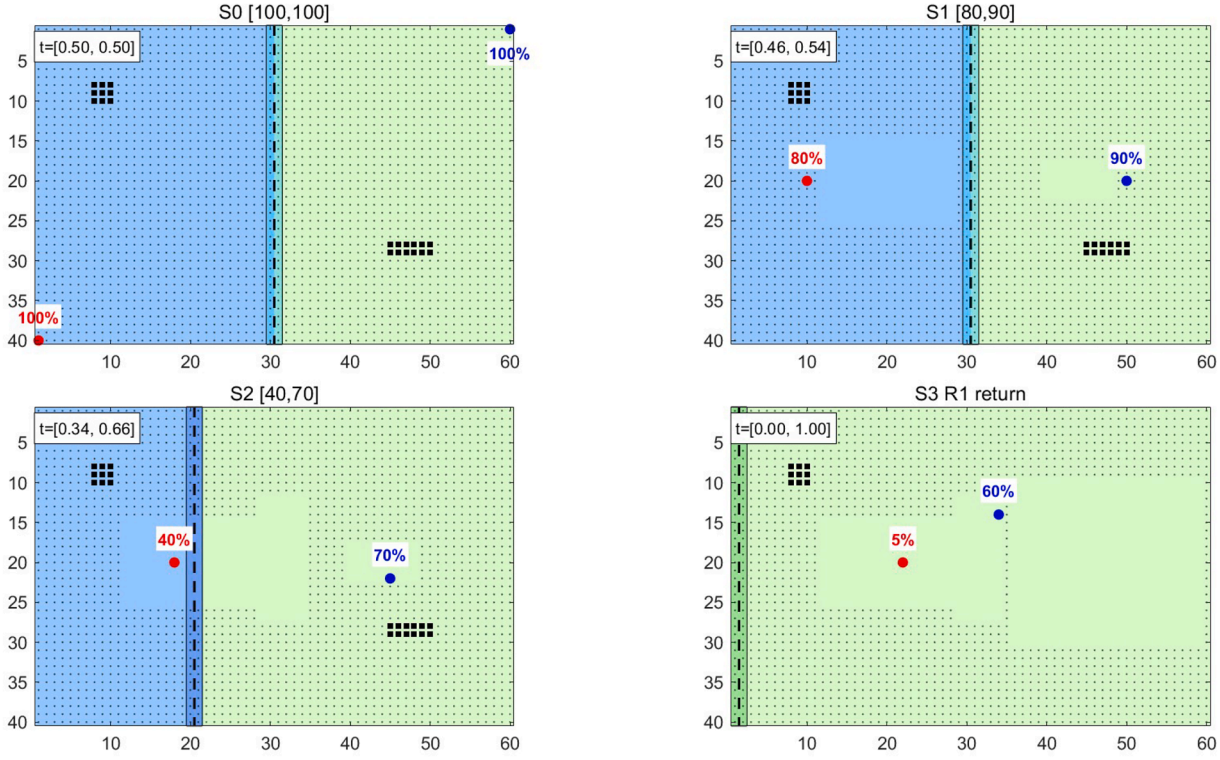


Fig. 3. Simulated DTT task.

Algorithm 2 DTT (Dynamic task transfer with soft-band).

Maintain strip-wise unknown mass $\{U^{(k)}\}_{k=1}^{n}$ and current partition R_{t-1}
Compute remaining unknown ratio $\rho_U \leftarrow U_{\text{rem}}/U_{\text{init}}$
if $(t - t_{\text{last}}) \geq H$ **and** $\text{mod}(t, P) = 0$ **and** $\rho_U > \sigma_{\text{stop}}$ **then**
  Compute desired workload share for each robot: $t_i \leftarrow \frac{a_i b_i}{\sum_j a_j b_j}$
  Compute cumulative unknown mass $C(c) \leftarrow \frac{\sum_{k=1}^{c} U^{(k)}}{\sum_{k=1}^{n} U^{(k)}}$
  Find cut index $c \leftarrow \arg\min_c \left|C(c) - t_1\right|$
  Define soft band $\mathcal{B} \leftarrow \{k : |k - c| \leq b_0\}$
  Hard-assign strips $k < c - b_0$ to robot 1 and strips $k > c + b_0$ to robot 2
  for each strip $k \in \mathcal{B}$ **do**
    Assign k to $i(k) = \arg\min_i d_M(\text{center}(k), p_i)$
  end for
  Update partition R_t and effective loads $\hat{U}_i \leftarrow \sum_{k \in R_t(i)} U^{(k)}$
  Update $t_{\text{last}} \leftarrow t$
else
  Keep previous partition $R_t \leftarrow R_{t-1}$
end if
return R_t and $\{\hat{U}_i\}$

with explicit safety constraints as a high-level policy guide. In environments characterized by high uncertainty, this strategy establishes a stable policy mapping through offline training, enabling two robots to collaborate in exploration without requiring additional online learning during deployment. Additionally, the reward function in this method integrates energy consumption and the output of the previously proposed CP-AFC and DTT mechanisms, aiming to balance exploration efficiency and energy utilization while enforcing mission safety requirements.

In this method, the double Q-learning state s'_t is a nine-dimensional vector, while the action $a_t \in A$ is selected from the set of discrete actions A. The reward function is designed to complement the CP-AFC and DTT methods.

$$s'_t = \left[d_M,\ q_\theta,\ b_t,\ \widehat{\text{EV}}_t,\ \widehat{\text{OL}}_t,\ \rho_t,\ \zeta_t^{(1)},\ \zeta_t^{(2)},\ \zeta_t^{(3)}\right],$$
$$A = \{\text{exploration},\ \text{energy saving},\ \text{switch region},\ \text{assist teammate},\ \text{wander},\ \text{return_home}\}. \tag{18}$$

Since SDQ is implemented as tabular Double Q-learning for macro-action selection, it requires a finite state space; therefore, it relies on discretized features rather than continuous input. The state vector is intentionally kept low-dimensional and task-oriented, since SDQ does not output low-level motion commands and does not replace geometric planning. Instead, CP-AFC and DTT determine the structured candidate targets and coordination context, while the low-level planner enforces feasibility. Under this division of responsibility, SDQ only needs coarse indicators that capture the safety margin, local traversal difficulty, and the context of collaboration. Discretization may introduce information loss at bin boundaries; however, the selected features preserve an ordinal structure that is sufficient for macro decisions, and the safety mask further prevents unsafe actions even when state resolution is coarse.

In this context, the Manhattan distance from the robot to the currently selected CP-AFC representative is defined as $d_M = |\Delta x| + |\Delta y|$. $q_\theta \in \{0, \ldots, Q-1\}$ represents the current task direction of the teammate, while Q is set to 4, which corresponds to the four quadrants. $b_t \in [0, 100]$ is the percentage of battery, $\widehat{\text{EV}}_t$ is the discretized score of the best frontier cluster selected by CP-AFC. $\widehat{\text{OL}}_t \in \{0, 1, 2\}$ represents the risk of conflict in the path of the robot, where $\widehat{\text{OL}}_t = 0$ indicates no risk and $\widehat{\text{OL}}_t = 2$ indicates high risk. $\rho_t \in [0, 1]$ is the local obstacle density in a fixed window. The additional terms $\zeta_t^{(1)}$, $\zeta_t^{(2)}$, and $\zeta_t^{(3)}$ are contextual indicators. Specifically, $\zeta_t^{(1)}$ captures the robot's short-term task transitions, $\zeta_t^{(2)}$ serves as a discretized indicator of the teammate's coverage progress, and $\zeta_t^{(3)}$ acts as a stability marker to detect repeated pauses or abnormal behaviors in recent actions. Each parameter is uniformly discretized into a fixed number of levels using linear binning, which

provides a practical trade-off between representational resolution and the sample efficiency required by offline tabular learning.

Based on the state s'_t, the reward function r_t is defined as

$$r_t = w_c\,\Delta\mathrm{cov}_t - w_e\,R_e - w_o\,R_{\mathrm{overlap}} + w_a\,R_{\mathrm{eff}} - w_d\,R_{\mathrm{dead}} + w_r\,R_{\mathrm{return}} - w_w\,R_{\mathrm{wander}}. \tag{19}$$

where $\Delta\mathrm{cov}_t$ is the coverage gain,

$$\Delta\mathrm{cov}_t = \frac{Cov_t - Cov_{t\text{-}1}}{\mathrm{totalcells}}. \tag{20}$$

R_e is the energy reward, which can be expressed as

$$R_e = k_d \times S_{\mathrm{frontier}} + k_\theta \times T_{\mathrm{turns}}. \tag{21}$$

where k_d is the coefficient of energy consumption of the robot per step, S_{frontier} are the steps to the frontiers, k_θ is the energy consumption of each turn, and T_{turns} is the number of turns. R_{eff} indicates the efficient reallocation of robot tasks in the DTT approach.

$$\begin{aligned} R_{\mathrm{eff}} &= \mathbf{1}\{\Delta\mathrm{trans}_t < -\eta\}, \\ \Delta\mathrm{trans}_t &= \sum_{i=1}^{N}\left|\alpha_{i,t} - t_{i,t}\right| - \sum_{i=1}^{N}\left|\alpha_{i,t-1} - t_{i,t-1}\right|. \end{aligned} \tag{22}$$

In Eq. (22), $\Delta\mathrm{trans}_t$ quantifies the variation in task imbalance, where larger negative values indicate a greater improvement in workload balance. The variables $\alpha_{i,t}$ and $t_{i,t}$ are defined in Eq. (11), while η is a constant threshold. The additional terms R_{overlap}, R_{dead}, R_{return}, and R_{wander} in Eq. (19) are derived from the detection of overlap, dead ends, successful return, and wandering behaviors on the map.

To support reproducibility and to avoid arbitrary parameter choices, the reward weights in Eq. (19) are selected prior to the main comparative evaluation using controlled validation runs and then fixed across all experiments. The tuning strategy aims to keep the major reward components at comparable numerical scales under typical operating conditions so that learning is not dominated by a single term. Coverage gain and energy cost form the primary objective trade-off, while overlap, dead-end, and wandering penalties are introduced to suppress degraded behaviors that are particularly common in fragmented maps, such as redundant revisits, getting trapped in low-connectivity branches, and prolonged non-productive motion. Although multi-objective rewards can hinder convergence when components are poorly scaled, training stability is supported here by the discretized state-action space and by Double Q-learning, which reduces value overestimation and improves learning stability in practice. Furthermore, the deterministic safety mask mechanism described below restricts inadmissible actions during training and deployment, thereby reducing the effective decision space and mitigating unstable exploration caused by unsafe or infeasible macro-action choices.

After obtaining the state s'_t and the shaped reward r_t, Eq. (23) is applied to conduct offline double Q-learning training [van Hasselt et al., 2016].

$$\begin{aligned} y_t &= r_t + \gamma Q^B\Big(s'_{t+1}, \arg\max_{a'} Q^A(s'_{t+1}, a')\Big), \\ L_A &= \big(y_t - Q^A(s'_t, a_t)\big)^2. \end{aligned} \tag{23}$$

In the deployment stage, a safety policy function $\mu(s')$ is adopted to eliminate additional online learning overhead and to ensure stability in new or previously unknown environments.

$$\mu(s'_t) = \arg\max_{a\in A}\{\, M_t(a)\, Q(s'_t, a)\,\}. \tag{24}$$

where $M_t(a) \in \{0, 1\}$ implements a deterministic safety mask mechanism. The safety mask is not learned. It is hard-coded and computed online based on the current energy state, the return-feasibility evaluation produced by SRS, and basic planning feasibility checks. SDQ learns the Q-values and selects actions only among those that remain admissible after masking. Concretely, return feasibility has the highest priority. When SRS determines that a safe return cannot be guaranteed under conservative evaluation, the robot is forced to execute return_home and all other actions are prohibited by setting their masks to zero. When return remains feasible, return_home is always considered admissible. If conservative return planning fails to produce a feasible path, SRS activates its breadcrumb fallback to backtrack along previously verified corridors. For exploration and energy saving, the action is masked if there is no reachable frontier target under the current admissible region constraints or if executing the action would reduce the remaining energy below the required safety margin that guarantees a feasible return. For switch region and assist teammate, the action is masked when the DTT is within its cooling period to prevent oscillatory switching, and it is also masked when the low-level planner cannot reach the target region or the assist target or when the remaining energy cannot support the motion while preserving the return feasibility. The wander action is masked whenever return is unsafe, and it is enabled only as a last-resort recovery behavior when recent state indicators detect stagnation or repeated non-productive motion, and no other productive admissible action is available.

This method prevents robots from performing unsafe or meaningless behaviors, reduces the likelihood of conflicts, and ensures that when the energy safety mechanism is triggered, the robot can safely return to its charging station. Algorithm 3 summarizes the SDQ procedure.

Algorithm 3 SDQ (safety double Q-Learning).

Initialize Q^A, Q^B and replay buffer D
for episode = 1 to N **do**
  Reset environment; get state s'_0
  for $t = 0, 1, \ldots$ until terminal **do**
    Build safety mask $M_t(a)$ using the deterministic feasibility rules described above
    if rand $< \varepsilon$ **then**
      Select random a with $M_t(a) = 1$
    else
      Select $a = \arg\max_{M_t(a)=1} Q^A(s'_t, a)$
    end if
    Execute a; observe s'_{t+1} and shaped reward r_t
    Store (s'_t, a, r_t, s'_{t+1}) in D
    if readyToLearn() **then**
      Sample minibatch from D
      if rand < 0.5 **then**
        $a^* = \arg\max_{a'} Q^A(s', a')$; $y = r + \gamma Q^B(s', a^*)$; update Q^A
      else
        $a^* = \arg\max_{a'} Q^B(s', a')$; $y = r + \gamma Q^A(s', a^*)$; update Q^B
      end if
    end if
  end for
end for
Deployment: choose $\mu(s'_t) = \arg\max_{a\in A}\{M_t(a)\, Q(s'_t, a)\}$

3.4. Safe-return switch (SRS)

After completion of the CP-AFC, DTT, and SDQ algorithms, exploration of the unknown environment can be effectively accomplished. To ensure mission reliability while maximizing exploration coverage, the SRS was introduced, which continuously monitors the robot's return energy requirements rather than relying on a fixed battery threshold. This dynamic mechanism estimates the conservative energy cost of returning to the base station in real time and initiates a return when the remaining energy is insufficient. By combining conservative path planning with a deterministic breadcrumb fallback, the SRS provides a robust two-layer guarantee. This ensures that each robot can complete the entire cycle from exploration to safe return, even in fragmented and highly uncertain environments.

Eq. (25) calculates the basic return energy requirement of the robot from its current position to the base station (initial position).

$$E_{\text{return}}^{\text{safe}} = (1+u)(k_d \times L_{\text{return}} + k_\theta \times T_{\text{turn}}). \tag{25}$$

Therefore, the return path is planned on a conservative map M^{cons}, which treats all unknown cells as obstacles. To keep the energy estimate up-to-date during exploration, the return path and energy cost will be evaluated in real time at each step.

Here, u is the return margin to prevent accidents, L_{return} is the length of the return path, computed by the A* algorithm, and T_{turn} is the total turning effort. This project illustrates a fragmented and unknown area where blindly crossing might lead to unexpected obstacles, resulting in energy waste. Therefore, the return path is planned based on the conservative map M^{cons} that treats all unknown cells as obstacles.

Based on the estimation of the return energy, Eq. (26) further defines the trigger condition of the SRS.

$$\text{Return} = \left(E_{\text{remain}} \le E_{\text{return}}^{\text{safe}}\right) \vee \left(\text{Cov}_t \ge \text{Cov}_{\min}\right) \vee \texttt{wander}. \tag{26}$$

where E_{remain} is the remaining energy of the robot, $Cov_{\min}$ is the minimum coverage threshold required for exploration, and wander is a Boolean flag that becomes true if the robot encounters a dead end on the map. The equation indicates that when the robot detects that its energy is below the required return energy, meets the minimum exploration requirements, or cannot find a path to any unknown areas, it will immediately stop its mission and return to its base station, and the other robot will take over its exploration mission.

To avoid extreme situations beyond static simulations where the conservative path planning algorithm M^{cons} fails to find a return path, the system will transition to a deterministic breadcrumb backtracking approach. The $\mathcal{B} = q_1, q_2, \ldots, q_K$ is used to denote the sequence of waypoints visited by the robot during exploration. The relaxed map M^{perm} is then constructed by temporarily marking all breadcrumb cells in $\mathcal{B}$ as traversable. The robot then re-plans a return path to the base station on this relaxed map M^{perm}, ensuring that a valid path can be found by leveraging the known visited cells. (Please note that in the simulated situation, a return route always exists in M^{cons}, thus the breadcrumb backtracking approach will remain inactive and will not affect any reported results.)

Finally, during multi-robot exploration, the safe return strategy takes priority to prevent any robot from running out of power. If a returning robot i and an exploring robot j are about to move into the same cell, priority is given to robot i. Robot j will either wait or temporarily retreat, ensuring that the returning robot will not be delayed. In addition, if a robot returns to its base station, it will remain silent. Once all robots have returned to the base stations, the mission is marked complete.

4. Experiment results

4.1. Experimental setup

4.1.1. Environment setting

This study uses MATLAB and its reinforcement learning toolbox to model and simulate the entire algorithm. The robot exploration scenario is simulated using a discrete grid-based environment representation. This grid environment is modeled as a two-dimensional discrete state space, where each cell represents a possible location for the robot to move or explore. To simulate autonomous exploration of a completely unknown scenario, all cells in the grid map are initially unknown, meaning the robot can only gradually reveal the unknown areas through continuous movement and sensor data collection.

The location and number of obstacles in the simulated environment are randomly generated, but diversity and repeatability are ensured through parameter control. Specifically, we use predefined obstacle density ratios (10%, 20%, and 30%) to vary the complexity of the map, thereby controlling the fragmentation of the free space traversable. For each complexity coverage configuration, we generate 100 independent random map instances and run each method on the same set of instances while fixing the pseudo-random seed. This environment ensures that all compared methods face the same scenario, and the reported statistics reflect the robustness of the methods across a wide range of randomized post-disaster topologies, rather than relying on a single hand-drawn map.

Fig. 4 presents a visual representation of the experimental results. In panel (a), we observe the original unknown map, where all free cells are marked as unknown. Although obstacles are indicated, it is important to note that the robots are unaware of their presence. Panel (b) illustrates the exploration areas covered by the two robots after they have completed their exploration. Panel (c) depicts the paths taken by the robots during the exploration. Finally, panel (d) shows the return paths of the two robots after finishing their exploration.

4.1.2. Robot setting

In this paper, two robots are placed in opposite corners of a map for exploration. We adopt a two-robot setting as a deliberate experimental choice rather than a restriction of the proposed framework. Two robots constitute the minimal team size that already exposes the key multi-robot phenomena targeted in this work, including frontier contention, congestion near narrow passages, and collaboration under asymmetric battery budgets. This controlled setting also enables fair and interpretable comparisons with existing baselines under identical randomized maps and communication settings, while avoiding confounding effects that arise with larger teams in highly fragmented layouts, such as severe corridor congestion and a rapidly expanded space of initial configurations. Importantly, CERES-Q is not inherently limited to two robots. CP-AFC can assign frontier ownership over a general set of robot identifiers, DTT can generalize from bilateral transfer to multi-region workload balancing using region-level unknown-rate and per-robot energy indicators, SDQ remains decentralized and can incorporate compact aggregated team summaries for decision making, and SRS is enforced per robot and thus scales directly. A systematic evaluation of performance versus team size is left for future work.

The robots' operating space is discretized into four basic motion directions: up, down, left, and right. The robots' perception is locally bounded; at each step, they update the occupancy of cells within their limited surrounding area, and their perception is obstructed by obstacles. Each robot has a limited energy budget that is consumed through movement and rotation, resulting in an energy-sensitive autonomy model. Battery status, pose, and mission objectives are shared between the robots. The robots periodically synchronize map data and maintain a shared global map.

In this work, the energy cost is modeled as

$$E_{\text{cost}} = pathlength + k_\theta \cdot turns, \tag{27}$$

where *pathlength* is the total traveled distance, *turns* is the number of turning maneuvers, and k_θ is the additional energy penalty associated with turning.

Based on experimental studies of skid-steer tracked rescue robots operating on rigid ground surfaces [46], the turning cost is typically equivalent to the energy required to travel between 0.15 and 0.6 meters in a straight line; therefore, we adopt a conservative mid-range value and set $k_\theta = 0.3$ in this study.

4.1.3. Baseline implementation details.

For reproducibility and fairness, we briefly summarize the implementation of the primary baseline Co-explore [47] used in our simulator. Co-explore extracts frontier clusters from the current occupancy grid and evaluates candidate targets by combining multiple normalized rankings, including travel distance to the candidate frontier and the utility of the frontier cluster. These rankings are fused into an assignment cost, and a one-to-one target allocation is produced using the Hungarian algorithm. In our implementation, Co-explore shares the same map update and synchronization schedule as CERES-Q, and all assigned targets

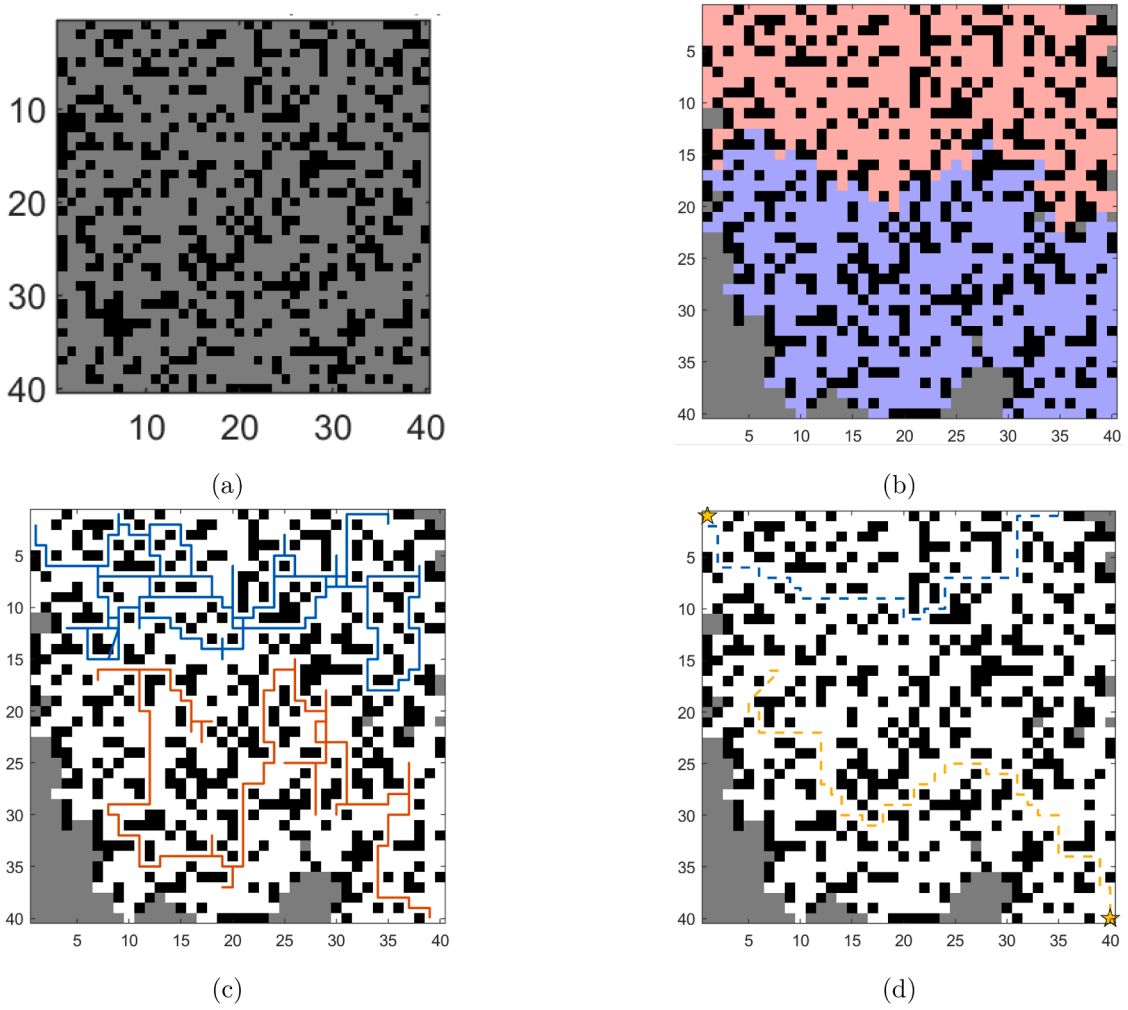


Fig. 4. Robot exploration sample. (a) Original unknown map. (b) Exploration map. (c) Exploration path. (d) Return path.

are executed by the same low-level A* planner on the current traversable grid, ensuring that performance differences reflect high-level coordination and decision logic rather than different motion planners.

4.2. Data analysis

4.2.1. Comparison with baseline methods at different complexities

In the data comparison, we will use the multi-robot BFS (Breadth-First Search), Boustrophedon, and the Co-explore algorithm proposed in recent years as the baseline for comparison. The results are shown in Table 2, Table 3, and Table 4.

Tables 2–4 reports a detailed quantitative comparison of the four methods at three levels of map fragmentation (10%, 20%, and 30%) and multiple coverage thresholds (75% – 95%). For each setting, we report the success rate (more than 100 trials) and the mean ± standard deviation of the path length, energy cost, number of turns, and achieved coverage. To quantify whether the differences are statistically significant, we also report the p-values of the Welch t-test for each baseline metric against CERES-Q. The symbol '-' indicates that no valid statistical comparison can be performed because the corresponding baseline does not produce successful trials under that setting.

In terms of robustness, CERES-Q achieves a success rate of nearly 100% in the simplest regime (10% complexity with 75%- 85% coverage), which is comparable to or slightly better than Co-explore. As the coverage requirement becomes stricter and the environment becomes more fragmented, the advantage becomes more pronounced. For example, at 20% complexity with a 90% coverage requirement, CERES-Q maintains a success rate of 97% while Co-explore drops to 92%, and the remaining baselines degrade more sharply (BFS at 78%, Boustrophedon at 0%). In the most challenging setting (30% complexity with a 95% threshold), CERES-Q is the only method that achieves a non-zero success rate (27%), which directly explains why statistical tests for other methods are not applicable there.

Regarding efficiency, CERES-Q consistently yields shorter travel distances than baselines, and these reductions are typically statistically significant in fragmented regimes. For example, with 20% complexity and 90% coverage, CERES-Q reduces the mean path length from 263.4 (Co-explore) to 152.7, with a highly significant difference ($p = 1.7 \times 10^{-66}$). A similar pattern appears in energy cost once fragmentation begins to dominate traversal overhead: at 20% complexity and 80% coverage, CERES-Q consumes 201.1 energy units on average, compared to 236.2 for Co-explore and 319.4 for BFS, and the improvement over Co-explore is highly significant ($p < 0.01$). At the same time, we observe that Co-explore can exhibit lower energy cost in the easiest maps due to smoother trajectories and substantially fewer turns, which may reduce or even eliminate statistical separability in isolated low-fragmentation cases. Nevertheless, as complexity increases, this advantage diminishes, while CERES-Q maintains a shorter overall travel distance and a stronger completion robustness. Finally, CERES-Q often exceeds the minimum coverage requirement by a safe margin (e.g., 78.9% achieved at a

Table 2
Results at Complexity = 10.

Cov	Method	Success	Pathlength	p	Energy	p	Turns	p	Coverage	p
75	BFS	100.0%	208.0 ± 11.1	9.4e-73	258.8 ± 15.1	2.7e-38	169.4 ± 18.3	3.3e-23	75.1 ± 0.1	3.1e-30
75	Boustro	48.0%	338.1 ± 19.4	1.1e-86	409.5 ± 24.9	1.4e-79	237.9 ± 23.9	0.226	75.0 ± 0.0	4.1e-31
75	Co-explore	100.0%	141.8 ± 18.5	9.6e-17	161.0 ± 22.6	6.0e-07	64.0 ± 16.0	3.2e-63	75.2 ± 0.2	1.3e-29
75	CERES-Q	99.0%	114.5 ± 23.3	—	183.7 ± 37.3	—	230.8 ± 46.6	—	78.9 ± 2.2	—
80	BFS	99.0%	224.8 ± 11.7	9.5e-63	280.9 ± 16.0	3.4e-31	186.9 ± 17.9	1.2e-21	80.2 ± 0.1	4.4e-31
80	Boustro	12.0%	399.2 ± 13.7	2.0e-27	480.3 ± 13.7	5.9e-42	270.3 ± 17.3	0.089	80.0 ± 0.0	3.6e-32
80	Co-explore	100.0%	156.6 ± 18.0	5.9e-15	179.0 ± 21.8	9.5e-07	74.7 ± 14.4	2.8e-56	80.2 ± 0.2	8.7e-31
80	CERES-Q	98.0%	127.6 ± 28.1	—	204.8 ± 44.9	—	257.2 ± 56.2	—	83.9 ± 2.2	—
85	BFS	99.0%	238.5 ± 11.4	3.4e-62	298.1 ± 15.5	2.4e-30	198.5 ± 17.4	1.5e-24	85.2 ± 0.1	9.7e-33
85	Boustro	1.0%	400.5 ± 0.0	—	481.8 ± 0.0	—	271.0 ± 0.0	—	85.0 ± 0.0	—
85	Co-explore	100.0%	175.7 ± 21.0	2.4e-20	200.9 ± 25.7	1.4e-04	84.0 ± 17.9	1.6e-58	85.1 ± 0.1	5.8e-33
85	CERES-Q	96.0%	138.1 ± 28.4	—	221.7 ± 45.4	—	278.5 ± 56.9	—	88.5 ± 1.8	—
90	BFS	94.0%	254.8 ± 12.9	5.1e-66	318.3 ± 18.1	1.9e-31	211.8 ± 20.8	9.1e-28	90.1 ± 0.1	2.5e-32
90	Boustro	0.0%	-	-	-	-	-	-	-	-
90	Co-explore	100.0%	202.9 ± 24.7	3.4e-31	233.3 ± 30.0	0.269	101.3 ± 19.7	3.3e-60	90.1 ± 0.1	8.5e-33
90	CERES-Q	96.0%	149.3 ± 28.4	—	239.4 ± 45.5	—	300.6 ± 57.0	—	93.0 ± 1.6	—
95	BFS	96.0%	275.5 ± 15.7	1.8e-54	344.3 ± 21.3	6.4e-18	229.2 ± 23.1	3.6e-35	95.2 ± 0.1	1.6e-26
95	Boustro	0.0%	-	-	-	-	-	-	-	-
95	Co-explore	98.0%	256.8 ± 28.6	6.2e-42	297.1 ± 34.7	0.065	134.4 ± 22.3	8.6e-58	95.1 ± 0.1	2.1e-28
95	CERES-Q	95.0%	177.7 ± 33.0	—	285.1 ± 52.9	—	358.0 ± 66.3	—	96.6 ± 1.0	—

Table 3
Results at Complexity = 20.

Cov	Method	Success	Pathlength	p	Energy	p	Turns	p	Coverage	p
75	BFS	84.0%	224.2 ± 17.6	1.1e-72	293.7 ± 23.4	1.4e-46	231.8 ± 21.9	0.933	75.1 ± 0.1	5.1e-34
75	Boustro	0.0%	-	-	-	-	-	-	-	-
75	Co-explore	100.0%	173.4 ± 22.4	2.4e-37	210.3 ± 28.7	2.5e-06	123.3 ± 23.4	5.4e-36	75.2 ± 0.2	8.9e-34
75	CERES-Q	99.0%	114.6 ± 28.4	—	184.0 ± 45.5	—	231.3 ± 56.8	—	79.0 ± 2.1	—
80	BFS	82.0%	243.2 ± 16.8	2.7e-70	319.4 ± 22.7	4.2e-45	254.2 ± 22.3	0.834	80.2 ± 0.1	1.5e-32
80	Boustro	0.0%	-	-	-	-	-	-	-	-
80	Co-explore	97.0%	195.0 ± 24.5	1.7e-40	236.2 ± 30.6	2.2e-08	137.2 ± 23.0	2.2e-34	80.2 ± 0.2	1.0e-32
80	CERES-Q	99.0%	125.3 ± 31.5	—	201.1 ± 50.5	—	252.8 ± 63.2	—	84.0 ± 2.1	—
85	BFS	87.0%	263.2 ± 19.2	2.3e-79	345.5 ± 24.9	1.2e-50	274.4 ± 21.6	0.129	85.2 ± 0.1	3.7e-34
85	Boustro	0.0%	-	-	-	-	-	-	-	-
85	Co-explore	92.0%	225.3 ± 26.1	4.3e-52	273.9 ± 32.9	1.7e-14	162.1 ± 24.9	6.7e-41	85.1 ± 0.1	1.3e-34
85	CERES-Q	98.0%	140.9 ± 28.4	—	226.1 ± 45.3	—	283.9 ± 56.5	—	88.6 ± 1.8	—
90	BFS	78.0%	280.6 ± 17.6	1.4e-77	367.9 ± 22.3	8.2e-49	290.9 ± 19.4	0.011	90.1 ± 0.2	7.8e-29
90	Boustro	0.0%	-	-	-	-	-	-	-	-
90	Co-explore	92.0%	263.4 ± 26.4	1.7e-66	320.8 ± 33.5	1.5e-26	191.3 ± 26.0	2.2e-36	90.1 ± 0.1	2.6e-29
90	CERES-Q	97.0%	152.7 ± 29.7	—	245.0 ± 47.6	—	307.7 ± 59.6	—	92.8 ± 1.6	—
95	BFS	58.0%	300.3 ± 14.5	3.0e-71	393.4 ± 18.9	1.4e-41	310.3 ± 17.6	1.8e-10	95.1 ± 0.1	7.1e-32
95	Boustro	0.0%	-	-	-	-	-	-	-	-
95	Co-explore	83.0%	339.4 ± 32.0	3.3e-79	414.7 ± 40.1	1.6e-46	251.0 ± 30.7	4.0e-31	95.1 ± 0.1	1.0e-32
95	CERES-Q	97.0%	177.1 ± 30.6	—	284.0 ± 48.9	—	356.6 ± 61.3	—	96.8 ± 0.9	—

75% threshold), whereas the baselines typically terminate closer to the threshold (75.1%–75.2%), improving reliability against stochastic variation near the stopping condition.

In general, these results demonstrate that the proposed system achieves a superior balance of robustness and efficiency. Specifically, it combines the stable and conflict-averse frontier clustering of CP-AFC, DTT's adaptive workload balancing by energy share, SDQ's safety-aware high-level decision-making, and the conservative return-feasibility guarantees of SRS. This synergy explains why our method achieves consistently high success rates across the tested settings and is the only approach that attains a non-zero success rate under the strictest coverage demand. At the same time, CERES-Q consistently produces shorter travel distances than the baselines and maintains competitive energy consumption, especially in regimes of moderate to high difficulty where fragmentation dominates the traversal cost. A key observation from Tables 2–4 is that CERES-Q often exhibits a higher number of turns than Co-explore. We hypothesize that this primarily reflects deliberate local maneuvering in cluttered, topologically constrained spaces rather than inefficient global motion. In fragmented maps, progress is frequently hindered by narrow passages and low-connectivity branches, which require frequent local reorientation to navigate bottlenecks and retreat from dead ends without committing to large detours. This behavior can incur extra local turning costs and may reduce efficiency in the simplest settings where smooth paths dominate; however, it becomes beneficial as fragmentation increases because it reduces large-scale backtracking and lowers the marginal travel required to acquire the remaining hard-to-reach cells near the target threshold. This hypothesis aligns with the statistics in Tables 2–4, where CERES-Q sustains higher success rates and shorter travel distances in high-complexity and high-coverage regimes despite a higher turning count.

In Figs. 5 and Fig. 6, we further analyze the exploration efficiency of CERES-Q. Fig. 5 shows the efficiency threshold curve, and Fig. 6 shows the energy coverage scatter plot, showing the coverage (x) and energy (y) achieved for each successful run. The broken line in the efficiency curve represents the threshold for unsuccessful trials in the baseline. The efficiency of each method could be calculated using Eq. (28).

$$\text{Efficiency} = \frac{\text{coverage}}{\text{total path length}}. \tag{28}$$

As shown in Fig. 5, at low complexity (10%), the co-exploration method performs best in the range 75% - 85%. This is because co-exploration has relatively few inflection points at low complexity. However, as the coverage threshold increases, its performance curve rises rapidly, while CERES-Q's performance increases more gradually. CERES-Q surpasses Co-explore when the coverage threshold reaches 95%. The other two baseline methods are relatively inefficient. At medium complexity (20%), CERES-Q shows the lowest performance

Table 4
Results at Complexity = 30.

Cov	Method	Success	Pathlength	p	Energy	p	Turns	p	Coverage	p
75	BFS	42.0%	252.9 ± 20.6	1.4e-27	341.4 ± 28.5	3.6e-12	295.2 ± 28.9	1.8e-06	75.2 ± 0.1	2.2e-18
75	Boustro	0.0%	-	-	-	-	-	-	-	-
75	Co-explore	54.0%	242.2 ± 36.2	1.1e-18	307.4 ± 47.5	0.001	217.4 ± 39.6	3.3e-22	75.1 ± 0.1	1.1e-18
75	CERES-Q	66.0%	171.9 ± 35.9	—	275.6 ± 57.5	—	345.8 ± 72.0	—	76.6 ± 0.9	—
80	BFS	38.0%	268.4 ± 16.3	2.0e-19	360.3 ± 21.7	1.6e-06	306.1 ± 21.1	6.3e-10	80.1 ± 0.1	8.4e-12
80	Boustro	0.0%	-	-	-	-	-	-	-	-
80	Co-explore	41.0%	268.1 ± 38.6	1.9e-14	338.5 ± 50.1	0.02	234.6 ± 40.8	8.8e-20	80.1 ± 0.1	9.6e-12
80	CERES-Q	48.0%	193.7 ± 36.8	—	311.0 ± 59.5	—	391.2 ± 75.6	—	81.6 ± 1.1	—
85	BFS	22.0%	283.2 ± 12.5	2.0e-13	381.0 ± 15.2	0.003	326.0 ± 14.2	2.0e-10	85.1 ± 0.1	4.6e-13
85	Boustro	0.0%	-	-	-	-	-	-	-	-
85	Co-explore	44.0%	312.3 ± 27.7	5.7e-20	395.7 ± 34.6	8.9e-05	277.9 ± 27.9	6.7e-16	85.1 ± 0.1	9.2e-13
85	CERES-Q	49.0%	215.8 ± 46.8	—	346.1 ± 75.1	—	434.4 ± 94.2	—	86.2 ± 0.8	—
90	BFS	3.0%	302.2 ± 18.6	0.074	402.0 ± 18.2	0.085	332.7 ± 4.7	4.0e-13	90.1 ± 0.0	1.2e-08
90	Boustro	0.0%	-	-	-	-	-	-	-	-
90	Co-explore	16.0%	341.1 ± 18.6	1.9e-08	427.5 ± 21.7	0.61	288.1 ± 16.5	1.1e-15	90.1 ± 0.1	1.5e-08
90	CERES-Q	34.0%	271.3 ± 53.3	—	435.6 ± 86.1	—	547.6 ± 109.3	—	91.0 ± 0.7	—
95	BFS	0.0%	-	-	-	-	-	-	-	-
95	Boustro	0.0%	-	-	-	-	-	-	-	-
95	Co-explore	0.0%	-	-	-	-	-	-	-	-
95	CERES-Q	27.0%	303.6 ± 43.7	—	486.9 ± 70.0	—	611.1 ± 87.5	—	95.6 ± 0.5	—

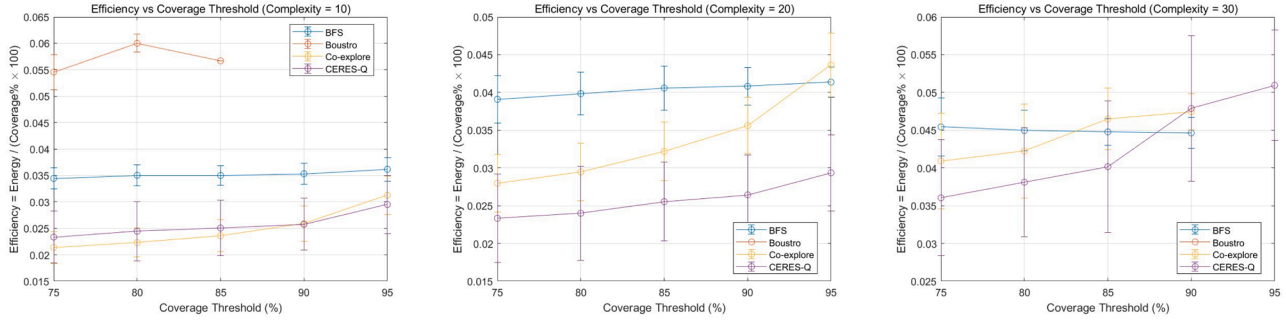


Fig. 5. Efficiency-coverage threshold curves at three map complexities (10%, 20%, 30%).

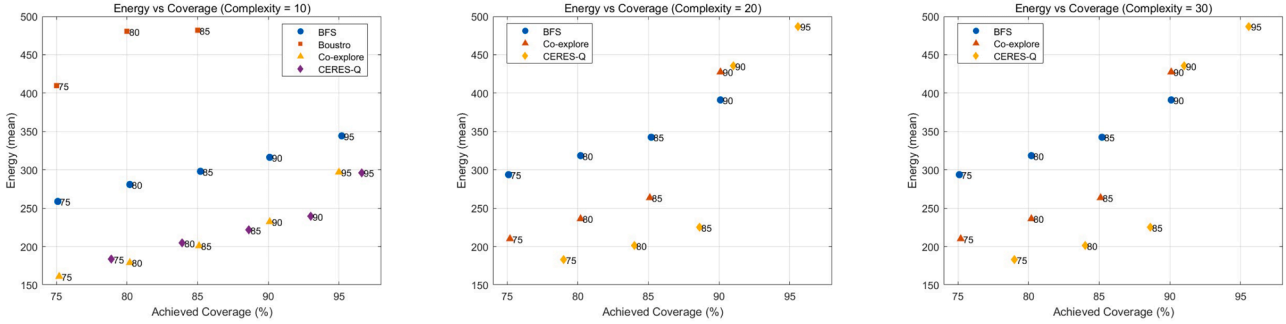


Fig. 6. Energy-coverage scatter plots for successful runs. Each point shows the achieved coverage and total energy. CERES-Q concentrates on the lower-left (higher coverage at lower energy), especially as complexity and coverage thresholds increase.

across all thresholds, while Co-explore's performance decreases more sharply with increasing coverage thresholds. Although the efficiency of BFS remains stable as the coverage threshold increases, its overall efficiency is not ideal. This indicates that CERES-Q maintains a significantly better distribution of unit coverage energy as the complexity of the task increases. At high complexity (30%), CERES-Q achieves the best performance within the range of complexity 75% to 85%. Importantly, in the most stringent setting (30% complexity and 95% threshold), all baseline methods fail, while CERES-Q still produces valid results, demonstrating feasibility through line continuity. This is further supported by error bars. Co-explore is compact for simple maps, but expands as the difficulty increases. In contrast, CERES-Q maintains a lower performance curve at higher thresholds with minimal variance increase, demonstrating robust efficiency under pressure.

Fig. 6 shows the shape of the curve from a distribution perspective. At each complexity level, CERES-Q's cloud tends to the lower left, indicating that CERES-Q achieves higher coverage with lower energy consumption, which improves task reliability under discretization and perceptual noise. In contrast, the Co-explore method clusters near the threshold at low energy in simple maps but drifts upward with increasing complexity and threshold, reflecting its steeper efficiency slope. The BFS method accumulates in a higher energy band and tends to end at the threshold. Meanwhile, the Boustro contribution success rate is too low to form stable clusters beyond the low threshold, consistent with its broken line. Taken together, Fig. 5 highlights which method scales better as the coverage requirement increases, while Fig. 6 represents which method has better energy coverage and safety margin. The analysis of these two different images shows a consistent conclusion: in the simplest

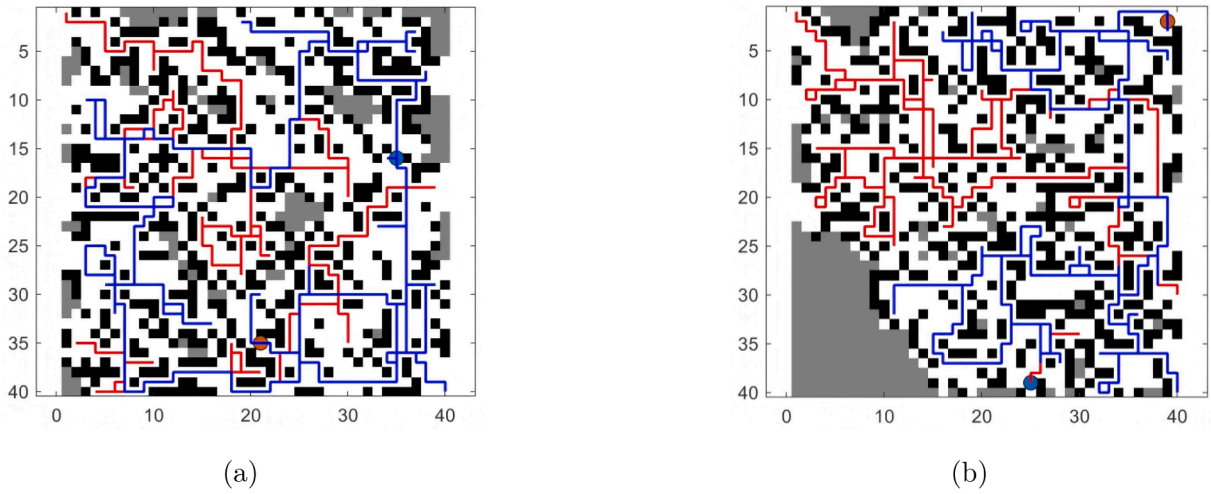


Fig. 7. Representative failure modes of Co-Explore in high-complexity fragmented scenarios. (a) Bottleneck oscillation. (b) Dead-end entrapment.

case, CERES-Q may be slightly lower than the marginal efficiency of the Co-explore method, but once the task becomes more complex, CERES-Q will show a lower and flatter efficiency trajectory than the three baseline methods. In summary, CERES-Q provides the most reliable and scalable performance in realistic, high-complexity, high-coverage environments, while maintaining competitive efficiency in simpler environments.

Beyond reporting success rates and statistical significance, we further examine typical failure patterns of baseline methods in highly fragmented environments, with a focus on Co-Explore [47]. Co-Explore combines multiple frontier-related rankings, including wavefront-based robot-to-frontier distances, distance-based robot ranking, and a frontier size/information-gain ranking, into a single cost matrix. It then computes assignments using the Hungarian algorithm at a fixed re-planning frequency, without waiting for robots to reach their assigned frontiers. Although this periodic reassignment is intended to continuously incorporate newly observed map information and to reduce oscillations via an implicit bias toward the currently pursued frontier, our high-complexity fragmented maps reveal two representative failure modes that can still emerge when the remaining frontiers are concentrated around choke points or low-connectivity branches.

Fig. 7 summarizes two representative failure modes of Co-Explore in highly complex, fragmented environments. First, Fig. 7(a) illustrates bottleneck-induced oscillation. Around a narrow chokepoint, multiple frontier candidates on adjacent branches often remain near-tied under Co-Explore's combined ranking (distance, rank, and size/information-gain cues), such that small local changes in robot position or newly revealed cells can repeatedly alter the optimal assignment. As a result, each robot may alternate between near-equivalent frontier candidates around the bottleneck, producing repeated local traversals with only limited net coverage progress.

Second, Fig. 7(b) presents a typical case of dead-end entrapment. Co-Explore drives the red robot into a low-connectivity branch in the upper-right region, where the route ahead terminates due to obstacles and forces backtracking through a constricted passage. In highly fragmented maps, such backtracking can incur a disproportionately high travel cost relative to the marginal coverage gain, especially when the remaining unexplored cells are located behind narrow passages and must be approached through long, circuitous routes. Under a limited time and energy budget, this leads to coverage stagnation and ultimately failure to complete the mission objective.

These representative failure modes provide a mechanistic explanation for the sharp increase in baseline efficiency cost as coverage approaches the target threshold (Figs. 5 and 6). In highly fragmented topologies, the remaining unexplored cells at high coverage levels are often located behind chokepoints or within low-connectivity regions, amplifying the impact of oscillations and dead-end backtracking on marginal coverage efficiency. In contrast, CERES-Q mitigates these behaviors through CP-AFC (stable frontier ownership and reduced contention), DTT (workload–battery-aware inter-region coordination), SRS (safety-triggered return-feasibility checks and return-aware overrides), and SDQ (macro-action selection that avoids unproductive switching), resulting in a smoother efficiency–coverage trend.

4.2.2. *Comparison with baseline methods at different energy level*

Table 5 shows the experimental results of dual robots in different environmental complexities (10%, 20%, and 30%) and with different initial battery levels (60%-60%, 40%-80%, and 20%-100%). In general, CERES-Q consistently outperforms the three baseline strategies in terms of coverage efficiency and energy utilization. As expected, higher complexity reduces the exploration coverage of all methods, but CERES-Q maintains a significant advantage. For example, under balanced battery conditions (60%-60%) with low complexity (10%), CERES-Q covers approximately 82.7% of the area, while Co-explore only achieves approximately 62.7%. Even with increasing complexity, both methods achieve a similar coverage of approximately 60%, significantly exceeding the other two baseline methods. However, CERES-Q maintains a low total path length and battery consumption, demonstrating that while the coverage of all methods decreases significantly with increasing task difficulty, CERES-Q remains competitive.

The advantages of our approach become even more evident under energy-imbalanced conditions. In the moderately imbalanced scenario, where the robot energy levels range from 40% to 80%, CERES-Q achieves coverage that is nearly as high as in balanced cases, while the performance of the other three baselines declines significantly. For example, with a complexity level of 20%, our strategy covers approximately 74.3% of the environment, with a total path length of about 165 ± 36 units. In contrast, Co-explore covers only around 62.3% of the area, with a travel distance of approximately 180 ± 68 units. This demonstrates that our robots are more efficient in exploring while traveling shorter distances. Furthermore, the data in Table 5 shows that our approach results in a more balanced distribution of tasks. In CERES-Q, the distance traveled by stronger robots is roughly 1.28 times that of weaker robots, which is the lowest ratio among those evaluated in CERES-Q. This suggests that our approach emphasizes the participation of lower-power robots in exploration, rather than overly relying on higher-power robots. A lower path-length ratio means that although high-battery robots take on more exploration tasks, low-battery robots still contribute significantly to the overall path taken. This strategy reduces the burden on high-battery robots and effectively prevents those

Table 5
Comparison of various methods under different complexity and power allocation.

Method	Energy	Cov	R1 path	R2 path	R1 energy	R2 energy	Total path	Total energy
Complexity = 10								
CERES-Q	20-100	75.71	33.07 ± 19.42	125.70 ± 66.94	44.51 ± 23.32	171.49 ± 89.24	158.78 ± 75.77	216.00 ± 92.79
CERES-Q	40-80	81.13	60.75 ± 37.72	113.40 ± 78.43	78.04 ± 48.64	116.79 ± 67.23	174.15 ± 61.18	194.73 ± 83.39
CERES-Q	60-60	82.74	82.89 ± 54.78	83.06 ± 57.59	108.56 ± 69.08	108.90 ± 69.74	165.96 ± 64.20	153.88 ± 98.06
Co-explore	20-100	60.03	18.97 ± 11.20	162.59 ± 75.05	25.94 ± 14.20	206.96 ± 109.95	181.57 ± 83.41	208.86 ± 110.86
Co-explore	40-80	65.42	59.69 ± 31.13	120.22 ± 79.40	77.88 ± 43.99	108.95 ± 55.57	179.90 ± 67.58	135.00 ± 71.44
Co-explore	60-60	80.20	88.02 ± 48.12	86.51 ± 53.51	113.16 ± 59.91	111.92 ± 63.45	174.52 ± 57.11	158.77 ± 87.08
BFS	20-100	79.36	41.20 ± 8.31	167.86 ± 67.53	55.25 ± 9.98	215.84 ± 75.97	209.05 ± 18.36	222.32 ± 77.63
BFS	40-80	79.53	62.76 ± 12.58	120.99 ± 18.43	72.70 ± 14.64	150.47 ± 27.33	183.74 ± 34.89	166.33 ± 31.16
BFS	60-60	78.20	88.07 ± 24.82	93.31 ± 33.39	115.58 ± 27.79	120.70 ± 33.72	181.39 ± 34.77	166.53 ± 43.41
Boustro	20-100	53.59	40.27 ± 22.42	137.41 ± 57.76	47.19 ± 25.50	168.07 ± 70.56	177.68 ± 58.07	215.27 ± 75.03
Boustro	40-80	56.33	82.10 ± 33.59	129.57 ± 59.04	97.83 ± 38.91	159.09 ± 72.29	211.67 ± 66.55	256.62 ± 82.18
Boustro	60-60	54.76	109.27 ± 48.63	103.23 ± 39.86	140.25 ± 61.01	132.02 ± 56.55	212.49 ± 63.70	192.42 ± 80.37
Complexity = 20								
CERES-Q	20-100	67.98	37.94 ± 14.80	115.62 ± 40.70	50.28 ± 18.09	159.28 ± 67.36	153.56 ± 46.23	166.68 ± 69.77
CERES-Q	40-80	74.25	73.72 ± 20.37	91.74 ± 26.10	92.01 ± 24.47	102.84 ± 27.93	165.47 ± 35.54	137.25 ± 36.91
CERES-Q	60-60	79.24	88.74 ± 22.14	88.06 ± 26.10	111.96 ± 27.24	110.82 ± 27.75	176.80 ± 25.14	157.26 ± 38.67
Co-explore	20-100	60.08	18.97 ± 9.44	162.59 ± 74.12	26.96 ± 11.09	204.57 ± 102.13	181.57 ± 83.41	206.47 ± 112.95
Co-explore	40-80	62.27	59.69 ± 25.44	120.22 ± 66.78	78.15 ± 28.54	108.30 ± 48.78	179.90 ± 67.58	135.09 ± 56.28
Co-explore	60-60	63.19	88.22 ± 49.82	93.62 ± 52.98	113.86 ± 59.56	118.45 ± 62.91	181.84 ± 55.81	164.98 ± 86.75
BFS	20-100	62.43	29.63 ± 10.36	148.19 ± 49.18	43.06 ± 11.90	193.47 ± 61.95	177.83 ± 44.52	198.47 ± 63.10
BFS	40-80	64.28	62.76 ± 12.58	120.99 ± 18.76	73.26 ± 15.66	150.72 ± 27.45	183.74 ± 34.89	166.76 ± 31.73
BFS	60-60	62.60	88.07 ± 24.82	93.31 ± 33.39	115.58 ± 27.79	120.70 ± 33.72	181.39 ± 34.77	166.63 ± 43.29
Boustro	20-100	40.94	24.69 ± 17.17	113.44 ± 73.33	31.03 ± 22.84	147.65 ± 95.11	138.13 ± 73.10	178.68 ± 97.34
Boustro	40-80	39.51	38.02 ± 30.17	94.59 ± 58.47	51.81 ± 41.76	123.37 ± 75.98	132.61 ± 55.01	171.43 ± 84.85
Boustro	60-60	31.82	40.51 ± 40.19	66.30 ± 44.10	51.39 ± 51.10	86.71 ± 57.66	106.80 ± 47.38	138.10 ± 77.04
Complexity = 30								
CERES-Q	20-100	56.21	33.07 ± 19.42	123.82 ± 66.11	44.59 ± 23.12	161.73 ± 85.11	156.89 ± 67.80	168.10 ± 88.92
CERES-Q	40-80	62.62	60.75 ± 37.72	113.40 ± 78.43	78.04 ± 48.64	116.79 ± 67.23	174.15 ± 61.18	139.33 ± 82.66
CERES-Q	60-60	60.66	82.89 ± 54.78	83.06 ± 57.59	108.56 ± 69.08	108.90 ± 69.74	165.96 ± 64.20	153.88 ± 98.06
Co-explore	20-100	59.10	18.29 ± 10.74	157.67 ± 85.47	25.54 ± 13.27	204.53 ± 111.37	175.97 ± 84.45	206.07 ± 112.64
Co-explore	40-80	62.93	58.28 ± 26.82	127.05 ± 81.11	78.77 ± 28.21	109.17 ± 48.65	185.32 ± 63.09	135.88 ± 56.44
Co-explore	60-60	60.87	88.02 ± 48.12	86.51 ± 53.51	113.27 ± 59.97	111.92 ± 63.45	174.52 ± 57.11	158.77 ± 87.08
BFS	20-100	50.20	22.47 ± 12.31	125.91 ± 66.37	36.10 ± 13.47	160.69 ± 76.86	148.38 ± 48.03	196.79 ± 78.10
BFS	40-80	44.68	43.72 ± 24.98	89.57 ± 24.60	59.20 ± 28.61	94.43 ± 33.41	133.30 ± 49.11	107.63 ± 56.43
BFS	60-60	45.70	67.38 ± 35.84	71.51 ± 47.83	81.05 ± 49.81	86.18 ± 47.83	138.89 ± 43.70	118.75 ± 70.51
Boustro	20-100	32.97	15.45 ± 14.43	101.13 ± 86.39	21.96 ± 21.06	117.69 ± 105.96	116.57 ± 56.34	119.65 ± 110.82
Boustro	40-80	28.94	24.02 ± 25.61	78.28 ± 71.24	32.01 ± 30.91	98.85 ± 95.38	102.30 ± 50.35	130.86 ± 100.22
Boustro	60-60	24.46	25.21 ± 28.70	60.24 ± 53.48	33.08 ± 33.65	85.45 ± 35.13	85.45 ± 35.13	118.53 ± 48.27

with a lower battery capacity from exiting exploration prematurely and becoming idle.

In extreme cases where one robot has a battery level of 20% and the other is at 100%, CERES-Q shows significantly better performance and fairness. For example, at 20% complexity, CERES-Q achieves an average coverage of 68.0% and a total path length of approximately 154 ± 46. This is significantly higher than Co-explore's coverage of 60.1%, which is accompanied by a longer path length of about 182 ± 83. In addition, compared to the BFS algorithm, the average coverage of our algorithm improves by about 5%, while the path length and power consumption are significantly shorter. Importantly, our approach utilizes nearly all available energy for both robots and effectively balances their workloads: the larger robot consumes only about 3.3 times the energy of the smaller robot, whereas the energy deviation in Co-explore is as high as approximately 8.6 times. This more equitable energy distribution maximizes the contribution of the weaker robot. However, it should be noted that the focus of CERES-Q on balance can slightly hinder peak coverage, particularly when a robot is significantly underpowered. For example, at the highest complexity level (30%), the average coverage of our approach is roughly 3% lower than that of Co-explore. However, this minor disadvantage is mitigated by the considerable improvements in efficiency and fairness of CERES-Q. In general, our approach reduces the total distance traveled by approximately 11% and maintains an energy usage ratio (R2:R1) of approximately 4:1, compared to the Co-explore efficiency ratio of 8.7:1. These results demonstrate that achieves superior coverage, lower path costs, and more balanced energy consumption, especially in challenging low-battery scenarios.

Fig. 8 illustrates the efficiency of the search path and the unit energy efficiency at complexity levels of 10%, 20%, and 30%, along with power ratios of 60%-60%, 40%-80%, and 20%-100%. The path efficiency histograms for the three complexity levels indicate that CERES-Q consistently achieves the highest path efficiency across all combinations. Regardless of terrain or power allocation, our approach consistently demonstrates superior path exploration efficiency. For example, on a map with complexity 20%, the search efficiency of CERES-Q in different power ratios is 0.448, 0.449, and 0.443, respectively, surpassing the best baseline method, Co-explore, which recorded 0.348, 0.346, and 0.330. This shows that CERES-Q effectively converts motion into coverage.

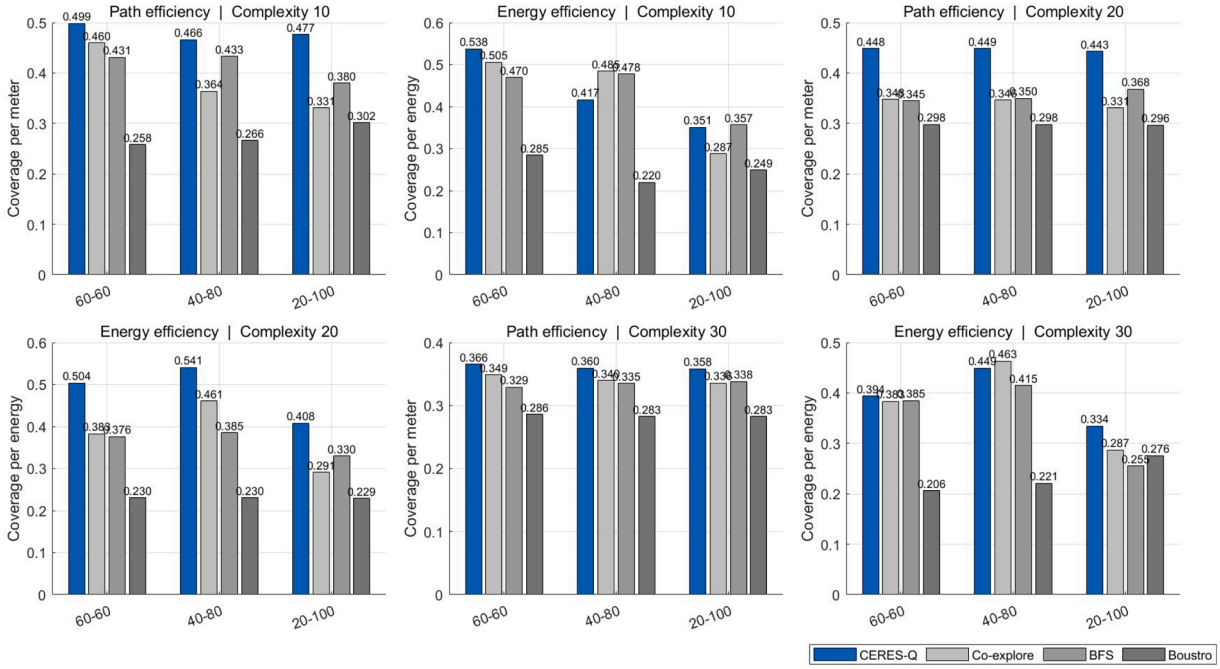


Fig. 8. Path efficiency and energy efficiency histograms across power splits.

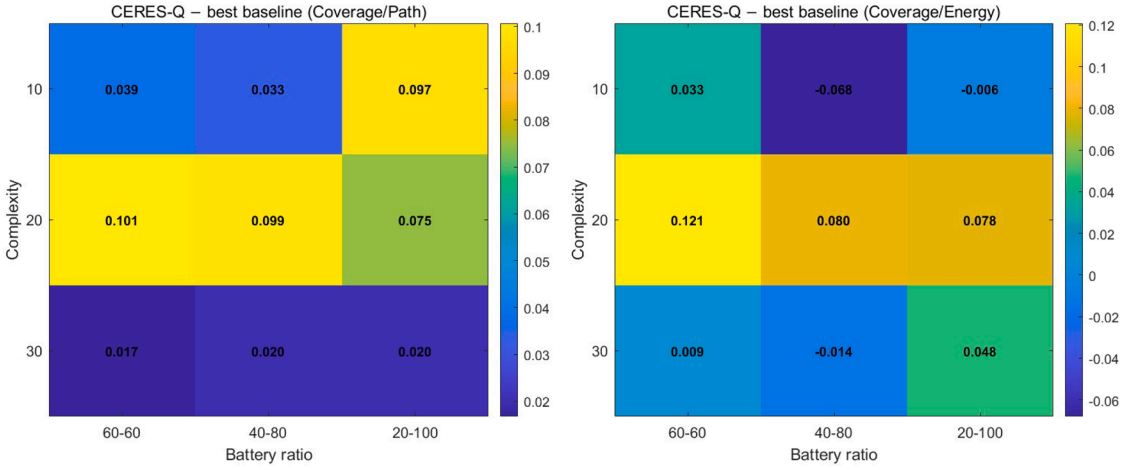


Fig. 9. Heat maps comparing the best-performing baseline vs. CERES-Q.

The three energy efficiency sub-graphs reveal that CERES-Q outperforms in energy efficiency in most scenarios, particularly when the two robots operate at significantly different power levels. In general, CERES-Q secured advantages in seven of nine different combinations of power and complexity, leading in the three power levels to a complexity of 20%. The only exceptions were at a complexity of 10% (robot power levels of 40% - 80%) and at a complexity of 30% (robot power levels of 40%-80%). However, in these cases, we still maintained the highest path efficiency, and the accompanying table (Table 5) demonstrates that CERES-Q achieved shorter total paths and greater coverage. These results indicate that our approach can occasionally trade lower energy consumption for increased coverage. Furthermore, at extreme power levels, such as a robot power ratio of 20% - 100%, CERES-Q showed superior path and energy efficiency, highlighting the effectiveness of our task allocation strategy at varying power levels.

Fig. 9 presents a heat map that compares CERES-Q with the top three baseline methods. These heat maps provide a more intuitive comparison between our approach and the baselines. The data indicate that CERES-Q generally outperforms the strongest baseline in translating robot exploration into coverage, and also excels in converting energy consumption into coverage in most instances. Coupled with the evidence from Table 5, which shows that CERES-Q achieves higher coverage and shorter total paths compared to other baseline methods, this strongly suggests that our approach more effectively transforms limited motion and energy into meaningful coverage, especially when the battery levels are low or unbalanced. This advantage becomes particularly prominent when there is a significant gap in battery levels.

4.2.3. *Ablation experiments*

To quantify the contribution of each major component in CERES-Q, we conduct an ablation study in a representative high-fragmentation setting with a complexity of 20% and a coverage requirement of 90%. Table 6 summarizes the results.

From Table 6, we find that the complete CERES-Q system achieves the highest reliability in this challenging regime, reaching a success rate of 100% and producing the lowest average path length and energy cost among the compared variants. Disabling the DTT method results in a substantial degradation in robustness, with the success rate dropping

Table 6
Ablation study at complexity 20% with a 90% coverage requirement.

Variant	Success(%)	Coverage(%)	Pathlength	Turns	Energy	Conflict
CERES-Q	100	96.15	193.78	378.9	307.45	19.9
w/o DTT	60	89.47	242.47	482.6	387.25	29.4
w/o CP-AFC	80	91.13	277.84	542.1	440.47	41.9
Greedy policy (w/o SDQ)	50	72.47	260.09	470.4	401.21	26.8

Table 7
Safe return analysis across complexities.

Complexity	Metric	Fixed-5%	Fixed-10%	Dynamic (Ours)
10	Return success rate (%)	96	100	100
10	Coverage (%, mean ± std)	90.8 ± 3.9	85.2 ± 3.6	92.5 ± 3.7
20	Return success rate (%)	88	95	100
20	Coverage (%, mean ± std)	88.1 ± 9.1	83.6 ± 8.3	89.4 ± 7.4
30	Return success rate (%)	76	92	100
30	Coverage (%, mean ± std)	72.3 ± 18.5	62.1 ± 17.3	76.9 ± 16.2

to 60% and the average cost increasing. This indicates that dynamic workload transfer is important for sustaining progress toward a stringent coverage target when the topology is highly fragmented and exploration workload becomes uneven across regions.

The replacement of CP-AFC with standard frontier processing also degrades performance. Although the success rate remains higher than the greedy-policy variant, the conflict metric increases markedly, and both the path length and the energy cost rise. This suggests that phase-consistent frontier clustering and temporary ownership cues are effective in suppressing contention-induced interference and reducing redundant traversal near bottlenecks.

Finally, substituting SDQ with a greedy high-level policy yields the lowest stability, reflected by a large variance in coverage and energy and a reduced success rate of 50%. This supports the role of safety-aware macro-action selection in preventing degenerate switching and inefficient motion patterns under strict objectives. The contribution of the safe-return component is further isolated and analyzed in the following subsection, where we compare fixed-threshold return strategies with our dynamic SRS mechanism.

4.2.4. Comparison between fixed return power and SRS

Table 7 compares three return strategies: two fixed battery thresholds (5% and 10%) and our energy-sensitive dynamic SRS. The metrics analyzed include the return success rate and the coverage of exploration. The fixed thresholds mimic the static return threshold method commonly used in the literature, whereas our strategy utilizes a pessimistic map to estimate the conservative energy required for docking in real-time, allowing for an adaptive return trigger. This difference becomes more pronounced as complexity increases because fixed-threshold rules implicitly assume that return feasibility is primarily a function of battery percentage and is only weakly affected by the evolving map topology. However, in highly fragmented environments, return trajectories are often constrained by narrow passages and low-connectivity branches, and the effective return cost can rise sharply due to detours, repeated turns, and newly discovered obstacles. As a result, a single fixed threshold can become either too late to guarantee a return in difficult maps or unnecessarily conservative in easier maps, whereas an online return-feasibility estimate can better align the trigger timing with the actual difficulty of the evolving environment.

In terms of robustness, the dynamic strategy achieves a 100% success rate for returns at all levels of complexity. In contrast, the fixed threshold approach presents a typical trade-off: when the return threshold is set at 5%, the robot may return too late and become lost, and this risk increases significantly with complexity (for example, it decreases to 76% at 30% complexity). On the other hand, setting the return threshold at 10% greatly improves the probability of a successful return (e.g., 92% in 30% complexity), but this comes at the cost of reduced exploration output. Our strategy addresses the trade-off between security coverage and safety by explicitly assessing the feasibility of returning to the current map and triggering a return only when a conservative return cost margin reaches a critical threshold. This allows SRS to remain non-intrusive in simpler environments while maintaining perfect return reliability as fragmentation increases.

Our strategy provides a relatively higher exploration coverage. At a complexity level of 10%, our coverage reaches 92.5%, surpassing both fixed thresholds (90.8% at 5% and 85.2% at 10%). This advantage increases as complexity increases. At a complexity level of 20%, coverage at the threshold of 10% drops to 83.6%, while our dynamic return approach and the threshold method of 5% maintain coverage close to 90%. At a complexity level of 30%, the results at the threshold 10% decrease significantly, and the coverage gap between our approach and the threshold method 5% widens. These trends suggest that online energy estimation and conservative re-planning can prevent premature termination without leading to return failures because the return decision is coupled to the evolving topological difficulty rather than a static percentage threshold. This effect is especially pronounced in fragmented, high-uncertainty maps, where the remaining return path can change substantially as exploration reveals new obstacles.

The results clearly show that the proposed dynamic SRS strikes an excellent balance between mission safety and coverage efficiency. In straightforward environments, it prevents unnecessary early return decisions that often occur with conservative fixed thresholds. In more complex environments, it ensures a high level of return reliability while significantly enhancing coverage. This indicates that, rather than relying on a one-size-fits-all fixed return threshold, our dynamic return-to-home approach effectively creates a return mechanism that adapts to the complexity of the environment.

5. Conclusion

This paper proposes a closed-loop multi-robot exploration framework called CERES-Q, designed specifically for fragmented and initially unknown environments. The system integrates four key components: Phase-Aware Front Clustering and Assignment (CP-AFC), Dynamic Task Transfer (DTT), Safe Return Switching (SRS) and Safe Double-Q Learning (SDQ). These modules together form a complete loop for perception, decision-making, planning, execution, and return. The design goal is to improve coverage robustness, team efficiency, and energy security, particularly in scenarios such as narrow passages, low-connectivity areas, and asymmetric power distribution onboard.

Empirical results at three levels of map complexity (10%, 20%, and 30%) and various battery allocation ratios (20%-100%, 40%-80% and 60%-60%) demonstrate that CERES-Q consistently achieves shorter total travel distances while reaching or exceeding the same coverage compared to classical baseline methods BFS, Boustro, and recently proposed co-exploration methods. CERES-Q excels in scenarios with high topology fragmentation and stringent task requirements, particularly when teams initially exhibit significant power imbalances. In such situations, the remaining unexplored areas are often located in narrow bottlenecks or poorly connected branches, making coordination stability and return feasibility constraints crucial. Under these conditions, CERES-Q demonstrates significantly greater improvements in robustness and efficiency across all battery configurations tested.

In the most challenging scenario (30% complexity, 95% coverage threshold), CERES-Q is the only method capable of maintaining a non-zero success rate. It also significantly reduces the path length while maintaining competitive or even higher coverage. Efficiency versus coverage curves and energy versus coverage scatter plots further demonstrate that as coverage requirements increase, the rate of efficiency decline flattens, and successful trials move toward achieving higher coverage with lower energy consumption. Even with unbalanced battery power, CERES-Q maintains efficiency and fairness, reflecting the anticipated coupling between workload allocation and energy-aware coordination. In particular, the dynamic safety return mechanism achieved a 100% docking success rate in all complex environments tested, eliminating the inherent trade-off between security and coverage in the fixed-threshold return strategy. In general, these results demonstrate that in complex and unknown environments, a robust balance between exploration efficiency, coverage integrity, and return safety can be achieved by combining macroscopic action selection with a structure-aware planner and explicit safety constraints.

Despite these advances, some limitations remain. Offline double-Q training remains time-consuming. Although our evaluation covers a large parameter-controlled randomized test set, we have not yet tested CERES-Q on standardized public benchmark maps for robot exploration, which limits broader cross-method comparisons. Our simulations also employ grid abstraction with static obstacles and periodic map synchronization under ideal communication, thereby simplifying uncertainties in perception and localization. In real post-disaster deployments, communication delays or dropouts can lead to stale shared maps and delayed coordination updates, reducing the effectiveness of repartitioning and frontier ownership, and increasing redundant exploration near bottlenecks. Although CERES-Q operates in a decentralized closed-loop system and can conservatively enforce safe returns using locally available information, a systematic sensitivity study across latency and packet-loss conditions is beyond the scope of this paper.

Scalability also depends on coordination overhead as the map size increases. CP-AFC performs frontier extraction on the grid and then applies phase-aware clustering and ownership assignment to the extracted frontier set. Its computational cost scales with the size of the map and the number of frontier cells and frontier groups. DTT is event-triggered and primarily updates region-level workload and energy metrics via boundary adjustments that can be maintained incrementally. In practice, the dominant costs typically lie in low-level planning and map updates, but larger maps and team sizes motivate incremental updates and hierarchical coordination.

Future work will focus on accelerating offline training through prioritized experience replay and transfer learning to warm-start policies in new environments. We will evaluate CERES-Q on public occupancy-grid benchmark maps and extend coordination toward asynchronous, delay, and dropout-aware communication. Key challenges for physical deployment include sensor noise and partial observability in mapping, localization drift affecting frontier and ownership estimates, mismatch between grid-based energy models and real motion costs, and non-holonomic constraints in continuous dynamics. Addressing these issues requires robust SLAM and uncertainty-aware mapping, planning, and control compatible with non-holonomic motion, and safety margins supported by online calibration. Finally, we plan to extend this framework to larger robot teams and continuous dynamics.

CRediT authorship contribution statement

Yao Xue: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization; **Chee Keong Tan:** Writing – review & editing, Supervision, Project administration; **Wai Peng Wong:** Writing – review & editing, Supervision.

Data availability

I have shared my data at the Attach File step

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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